PHYSICS

1) A charged conducting sphere has diameter 10m and surface charge density $\frac{\varepsilon_0}{4\pi}\frac{C}{m^2}$. Electric flux passing through its surface will be :-

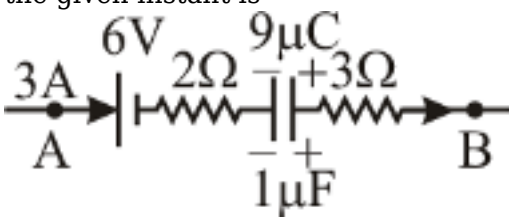
(1) 25 Nm^2/C

(2) 16×10^7 Nm^2 /C

(3) 10^6 Nm^2/C

(4) 2.2×10^8 Nm^2/C

2) The potential difference V_A–V_B between points A and B for the circuit segment shown in figure at the given instant is

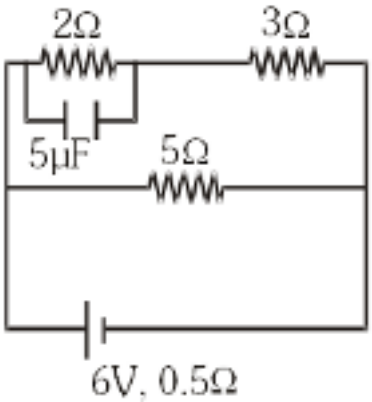


(1) 12 V

(2) –12 V

(3) 6 V

(4) –6 V



3) Find out amount of charge on capacitor shown in circuit.

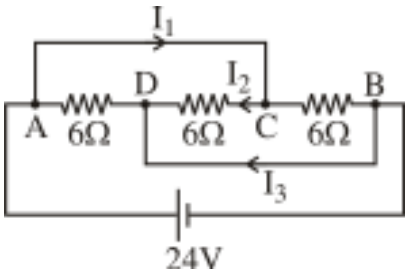
(1) 0μC

(2) 30μC

(3) 10μC

(4) 25μC

4) Which of the following options is incorrect regarding the below diagram ?



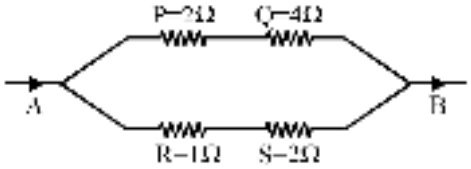
(1) I_1 = 8 Amp.

(2) I_2 = 4 Amp.

(3) I_1 = 0

(4) I_3 = 8 Amp.

5) Which of the four resistances P, Q, R and S generate the greatest amount of heat when a current



flows from A to B ?

(1) Q

(2) S

(3) P

(4) R

6) H^+, He^+ and O^{++} ions having same kinetic energy passes through a region of limited uniform magnetic field B directed perpendicular to the velocity of ions. The masses of the ions H^+, He^+ and O^{++} are respectively in the ratio 1 : 4 : 16. As a result:-
(A) H^+ ions will be deflected most
(B) O^{++} ions will be deflected least
(C) He^+ and O^{++} ions will suffer same deflection
(D) All ions will suffer the same deflection

(1) A, C

(2) B, C

(3) A, B, C

(4) A, B, C, D

7) A bar magnet of magnetic moment 'μ' is cut in three parts perpendicular to its length in ratio 2 : 4 : 6. Now all three parts combined in such a way that their axis are mutually perpendicular to each other then magnetic moment of system is :-

(1) $\sqrt{\frac{7}{18}}\mu$

(2) $\frac{\sqrt{7}}{6}\mu$

(3) $\frac{\mu}{12}$

(4) Zero

8) A toroidal winding has a total of 400 turns on a core with inner radius 8 cm and outer radius 10 cm. Calculate the magnitude of the magnetic field at a point midway between the inner and outer walls of the core when there is a current of 0.9 A is maintained in the windings.

(1) $4\pi \times 10^{-4}$ T

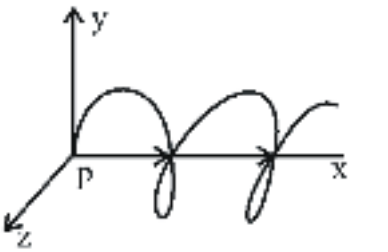
(2) $2\pi \times 10^{-2}$ T

(3) $3\pi \times 10^{-20}$ T

(4) $4\pi \times 10^{-10}$ T

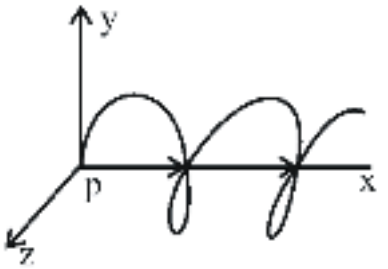
9) An electron is projected from origin with velocity $\left(3.2\hat{i}+\hat{j}\right)$ (m/s). A uniform magnetic field of $2\hat{k}$ tesla is present. The path followed by electron is correctly shown in
($m_e = 9 \times 10^{-31}$kg)

(1)



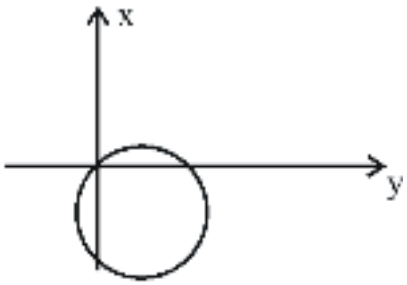
Helix on +ve side of z-axis

(2)



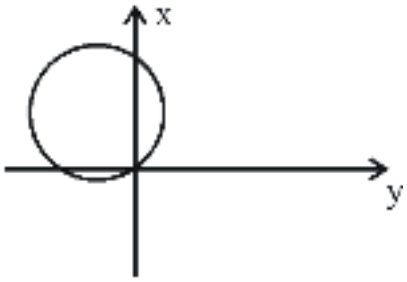
Helix on -ve side of z-axis

(3)



circle in xy plane

(4)



circle in xy plane

10) Two parallel, long wires carry currents i_1 and i_2 with $i_1 > i_2$. When the currents are in the same direction, the magnetic field at a point midway between the wires is 10 mT. If the direction of i_2 is reversed, the field becomes 30mT. The ratio i_1 / i_2 is

(1) 4

(2) 3

(3) 2

(4) 1

11) The magnetic field at a point X on the axis of a small bar magnet is equal to the field at a point Y on the equator of the same magnet. The ratio of the distances of point X and Y from the centre of the magnet is :-

(1) 2^{-3}

(2) $2^{-1/3}$

(3) 2^{3}

(4) $2^{1/3}$

12) A square frame of side 1 m carries a current I, produces a magnetic field B at its centre. The same current is passed through a circular coil having the same perimeter as the square. The magnetic field at the centre of the circular coil is B'. The ratio B/B' is :-

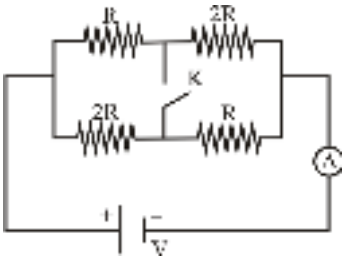
(1) $\frac{8}{\pi^2}$

(2) $\frac{8\sqrt{2}}{\pi^2}$

(3) $\frac{16}{\pi^2}$

(4) $\frac{16}{\sqrt{2}\pi^2}$

13) How will the reading of ammeter change if the key k is closed ?



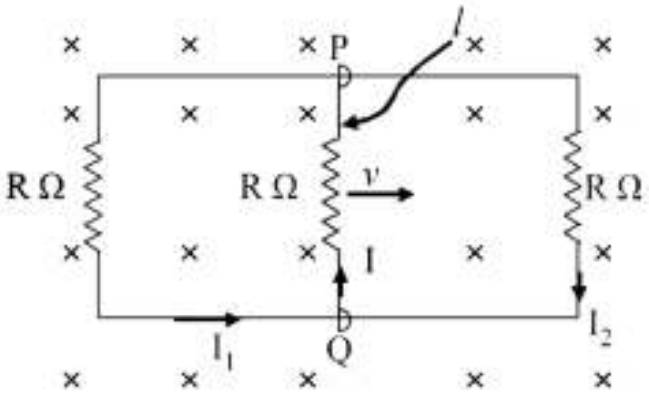
(1) Increase

(2) Decrease

(3) Remains same

(4) Information insufficient

14) A rectangular loop has a sliding connector PQ of length ℓ and resistance RΩ and it is moving with a speed v as shown. The set-up is placed in a uniform magnetic field which is into the plane of



the paper. The three currents I_1, I_2 and I are

(1) $I_1 = I_2 = \frac{Blv}{6R}, \quad I = \frac{Blv}{3R}$

(2) $I_1 = -I_2 = \frac{Blv}{R}, \quad I = \frac{2Blv}{R}$

(3) $I_1 = I_2 = \frac{Blv}{3R}, \quad I = \frac{2Blv}{3R}$

(4) $I_1 = I_2 = I = \frac{Blv}{R}$

15) Regarding self-inductance :
A : The self-inductance of the coil depends on its geometry.
B : Self-inductance does not depend on the permeability of the medium of core.
C : Self-induced e.m.f. opposes any change in the current in a circuit.
D : Self-inductance is electromagnetic analogue of mass in mechanics.
E : Work needs to be done against self-induced e.m.f. in establishing the current.
Choose the correct answer from the options given below :

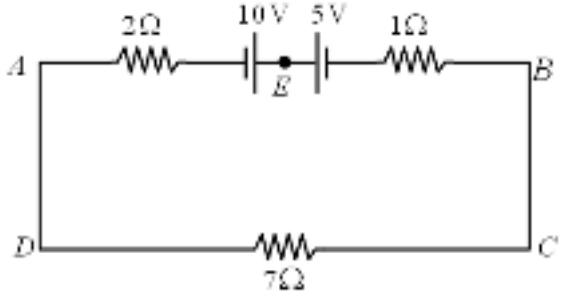
(1) A, B, C, D only

(2) A, C, D, E only

(3) A, B, C, E only

(4) B, C, D, E only

16) Which of the following option is NOT correct :-



(1) Current is 0.5 A from *A* to *B* through *E*

(2) $V_A - V_E = 9V$

(3) $V_E - V_B = 5.5V$

(4) 5W power is being supplied by 10V source

17) An armature coil consists of 30 turns of wire, each of area 0.05 m^2 and total resistance of 10Ω. It rotates in a magnetic field of 0.15 T at a constant frequency of $\frac{140}{\pi}$ Hz. Determine the value of maximum induced e.m.f. produced in the coil :

(1) 1 volt
(2) 500 volts
(3) 63 volts
(4) 43 volts

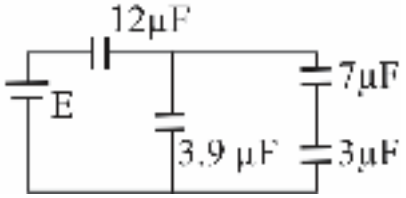
18) A rectangular loop of length 2.5 m and width 2 m is placed at 60° to a magnetic field of 4 T. The loop is removed from the field in 10 sec. The average emf induced in the loop during this time is : -

(1) – 2V
(2) + 2V
(3) + 1V
(4) – 1V

19) Two concentric rings of radii $R_1 = \sqrt{6}$ m and $R_2 = 4m$ are placed in y-z plane with their centres at origin. They have uniform charge –q and Q (where $Q = 2\sqrt{2}q$) on the inner and outer rings respectively. Consider the electrostatic potential to be zero at infinity. Then choose the correct option(s) :-
(A) Potential at origin is zero
(B) Field at origin is zero
(C) Field at x = 2 is zero
(D) None of these

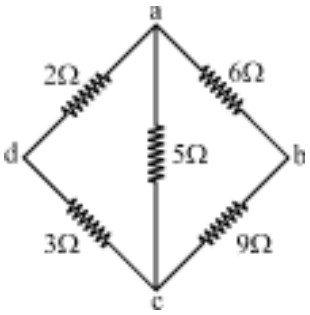
(1) A, B
(2) B, C
(3) A, C
(4) D

20) Four capacitors and a battery are connected as shown. The potential drop across the 7μF capacitor is 6V. Then find emf 'E' (in Volt) of the battery.



(1) 30 V
(2) 20 V
(3) 10 V
(4) 40 V

21) In the figure shown, the value of each resistance is marked there. Match the entries of Column-I



with the entries of Column-II.

Column-I		**Column-II**	
(A)	Resistance between a and b	(P)	$\frac{24}{5}\Omega$
(B)	Resistance between d and b	(Q)	$\frac{69}{35}\Omega$
(C)	Resistance between a and c	(R)	$\frac{138}{35}\Omega$
(D)	Resistance between d and c	(S)	$\frac{15}{7}\Omega$
		(T)	$\frac{153}{35}\Omega$

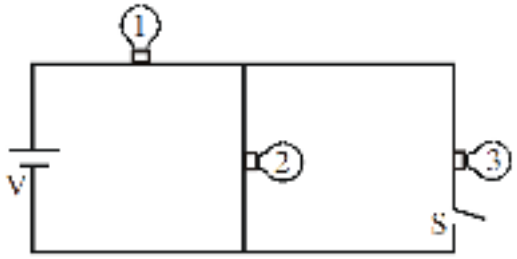
(1) A → R ; B → P ; C → S ; D → Q

(2) A → S ; B → P ; C → Q ; D → R

(3) A → Q ; B → S ; C → P ; D → R

(4) A → P ; B → Q ; C → R ; D → S

22) Figure shows 3 identical bulbs with switch S is open. The brightness of bulb 1 is B_1, bulb 2 is B_2 and net brightness of system is B_T. If switch is now closed, the brightness of :-



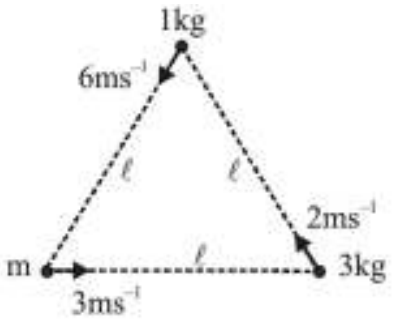
(1) B_1 remains same, B_2 increases and B_T increases.

(2) B_1 remains same, B_2 decreases & B_T decreases

(3) B_1 increases, B_2 decreases & B_T increases

(4) B_1 decreases, B_2 increases & B_T increases

23) Three particles of masses 1kg, m (kg) and 3kg are situated at the corners of an equilateral triangle move at speed 6 ms^{-1}, 3ms^{-1} and 2ms^{-1} respectively. Each particle maintains a direction towards the particle at the next corner symmetrically. If the velocity of centre of mass of the system



at this instant is zero then find the value of m.

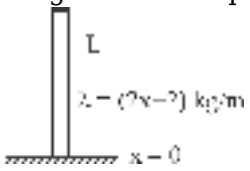
(1) 1 kg

(2) 2 kg

(3) 3 kg

(4) 4 kg

24) A vertical rod of length 1 m has a non-uniform linear mass density given by λ = (2x + 2)kg/m. Taking the ground as the reference level for gravitational potential energy, determine the gravitational potential energy of the rod.



(1) $\frac{5g}{3}$

(2) $\frac{2g}{3}$

(3) $\frac{4g}{3}$

(4) none

25) A bomb is projected with a speed of 20 m/s at an angle such that its horizontal range is 40 m. At the highest point of its trajectory, it explodes into two fragments of equal mass. Immediately after the explosion, one fragment comes to rest. Determine the horizontal distance from the point of projection to the point where the other fragment strikes the ground.

(1) 20 m

(2) 60 m

(3) 40 m

(4) 30 m

26) Two particles A and B are projected from ground with speed 2 m/s and $\frac{4}{\sqrt{3}}$ m/s from horizontal with angle 30° and 60° respectively. If seperation between A and B on ground is 10 m. Find relative angular velocity of B w.r.t A :

(1) 0.1 rad/s

(2) 0.2 rad/s

(3) 0.05 rad/s

(4) 1 rad/s

27) A vehicle is moving on a banked circular road. If the angle of banking is kept constant and the radius of curvature of the road is increased to four times its original value, determine the percentage increase in the maximum speed with which the vehicle can safely negotiate the curve.

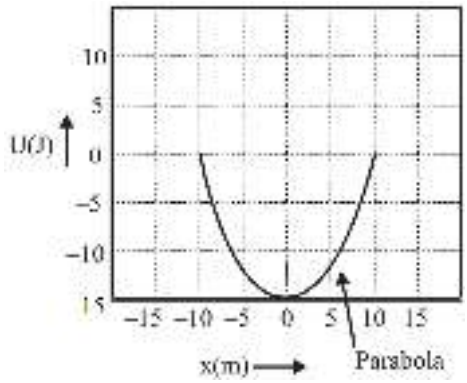
(1) 25%

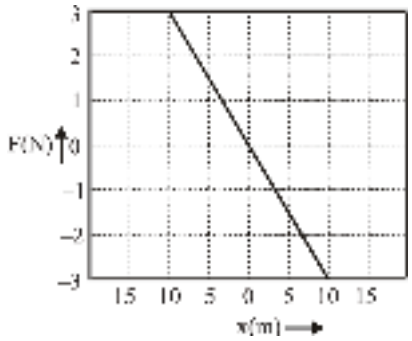
(2) 50%

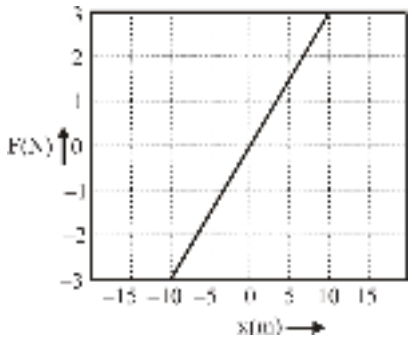
(3) 75%

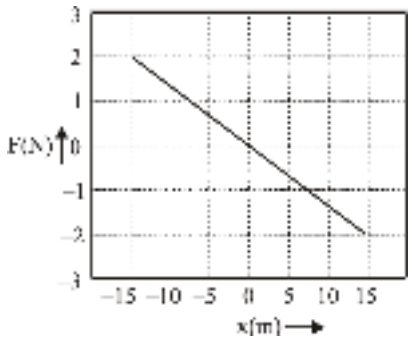
(4) 100%

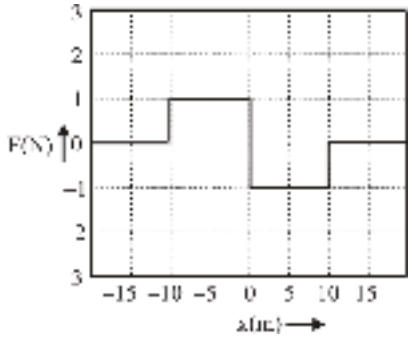
28) The graph of **potential energy (U)** versus **position (x)** for a particle moving along the **x-axis** is shown in the figure. No force other than the conservative force associated with the given potential energy acts on the particle. Which of the following graphs correctly represents the **force (F)** as a function of **position (x)** corresponding to the given potential energy graph?



(1)


(2)


(3)


(4) 

29) **Assertion :** No work is done by a force on an object when the point of application of the force does not move.
Reason : A small block of mass m is kept on a rough inclined surface of inclination α (with horizontal) fixed in an elevator. The elevator goes up with a uniform velocity u and the block does not slide on the wedge. Then the work done by the force of friction on the block in time t will be $mgut\sin^2\alpha$.

(1) Both Assertion and Reason are the true and Reason is a correct explanation of Assertion.

(2) Both Assertion and Reason are the true but Reason is not a correct explanation of Assertion.

(3) Assertion is true and Reason is false.

(4) Assertion is false and Reason is true.

30) A machine which is 75% efficient, uses 12 J of energy in lifting 1 kg mass through a certain distance. The mass is then allowed to fall through the same distance, the velocity at the end of its fall is :-

(1) $\sqrt{12}$ m/s

(2) $\sqrt{18}$ m/s

(3) $\sqrt{24}$ m/s

(4) $\sqrt{32}$ m/s

31) A uniform chain of mass m and length is held on a frictionless table in such a way that its $\frac{2\ell}{5}$ part is hanging below the edge of table. Find work done to pull the hanging part of chain on the table.

(1) $\frac{2mg\ell}{25}$

(2) $\frac{mg\ell}{50}$

(3) $\frac{mg\ell}{25}$

(4) $\frac{4mg\ell}{25}$

32) Two forces $\vec{F}_1 = (t^2-1)\hat{j}\,N$ and $\vec{F}_2 = \left[(1-t)\hat{i} + (t^2-1)\hat{j}\right]\,N$ are acting on a body. Determine the value of time (t) at which the body will be in equilibrium

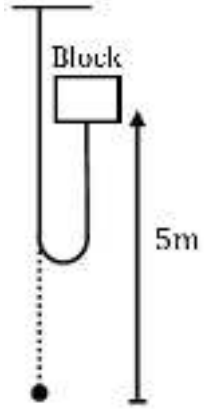
(1) 2s

(2) 1s

(3) 3s

(4) 4s

33) A 2 kg block is suspended by a string as shown in the figure. It is released from rest from a height of 5 m. Determine the impulse imparted by the string at the instant it just becomes taut.



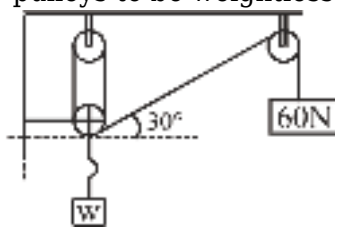
(1) 2 N-s

(2) 0.2 N-s

(3) 20 N-s

(4) 200 N-s

34) If the system shown in the figure is in equilibrium, then calculate the value of weight w. Assume pulleys to be weightless and frictionless :



(1) 60N

(2) 120N

(3) 150N

(4) 90N

35) A block of mass 5 kg is given horizontal velocity $(\hat{i}+\hat{j})$ m/sec on a rough horizontal surface. If coefficient of kinetic friction of surface is $\mu_k = 0.5$, then find the friction force vector acting on the block. (take g = 10 m/sec^2)

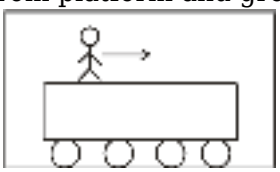
(1) $25(\hat{i}+\hat{j})$

(2) $-25(\hat{i}+\hat{j})$

(3) $\frac{-25}{\sqrt{2}}(\hat{i}+\hat{j})$

(4) $15\hat{i}+20\hat{j}$

36) A person 'A' is walking on a trolley with an acceleration of $2\hat{i}\ m/s^2$. The mass of the person is 50 kg. The trolley is being moved on a platform with velocity of $-3\hat{i}\ m/s$ and the platform is moving on ground with an acceleration of $-5\hat{i}\ m/s^2$. Then the pseudo forces on person when viewed from platform and ground are respectively :-



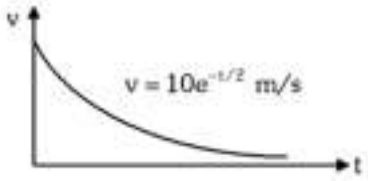
(1) $150\hat{i}$ N and $-250\hat{i}$ N

(2) $250\hat{i}$ N and zero N

(3) zero N and $-250\hat{i}$ N

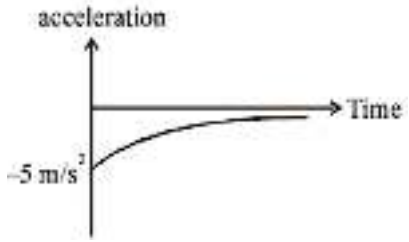
(4) $-250\hat{i}$ N and zero N

37) Velocity-time (v–t) graph of a particle moving in straight line is shown in the given figure.

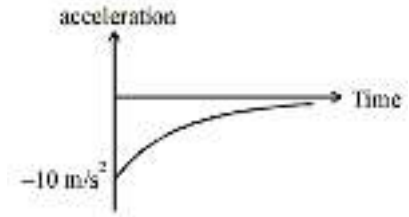


Which of the following graphs shows variation in acceleration of the particle with time?

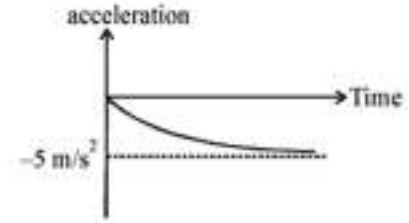
(1)



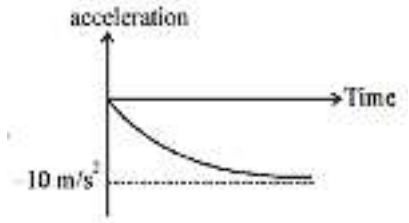
(2)



(3)



(4)



38) A particle is moving in x–y–plane at 2 m/s along x–axis. 2 seconds later, its velocity is 4 m/s in a

direction making 60° with positive x–axis. Its average acceleration for this period of motion is:-

(1) $\sqrt{5}$ m/s^2, along y–axis

(2) $\sqrt{3}$ m/s^2, along y–axis

(3) $\sqrt{5}$ m/s^2, along at 60° with positive x–axis

(4) 3 m/s^2, at 60° with positive x–axis.

39) A particle covers first one third distances with speed v_1 and remaining distance with speed v_2 then its average speed is :

(1) $\frac{3v_1v_2}{v_1+v_2}$

(2) $\frac{3v_1v_2}{2v_1+v_2}$

(3) $\frac{3v_1v_2}{v_1+2v_2}$

(4) Zero

40) Two tall buildings are 30m apart. The speed with which a ball must be thrown horizontally from a window 150 m above the ground in one building so that it enters a window 27.5 m from the ground in the other building is :

(1) 2ms^{-1}

(2) 6ms^{-1}

(3) 4ms^{-1}

(4) 8ms^{-1}

41) In a simple pendulum experiment, the time taken for 25 oscillations is measured to be 50 s when the length of the pendulum is 100 cm. If the least count of the stopwatch is 0.1 s and that of the meter scale is 0.1 cm, determine the maximum possible percentage error in the calculated value of g.

(1) 0.5 %

(2) 1 %

(3) 0.4 %

(4) 0.1 %

42)

The measured values of two physical quantities are :
x = 12.0 ± 0.6, y = 5.0 ± 0.2

Let P = xy, D = $\frac{x}{y}$, R = x + y, S = x – y.

Arrange the quantities P, D, R and S in the increasing order of their percentage errors.

(1) R < S < P = D

(2) R < S < P < D

(3) $R < P = D < S$

(4) $P = D < R < S$

43) If the energy (E), velocity (v) and force (F) be taken as the fundamental quantity, then the dimensions of mass will be :-

(1) Fv^{-2}

(2) Fv^{-1}

(3) Ev^{-2}

(4) Ev^{2}

44) The diameter of a spherical object is measured using a vernier calipers. The 10 vernier scale divisions (V.S.D.) are equal to 9 main scale divisions (M.S.D.), and the value of 1 M.S.D. is 0.1 cm. When the jaws of the vernier calipers are completely closed, the zero of the vernier scale is at x=0.1 cm, indicating a zero error. If the main scale reading for the diameter is M = 5.0 cm and the 8th vernier division coincides with a main scale division, determine the corrected diameter of the spherical object after applying the zero error correction.

(1) 5.18 cm

(2) 5.08 cm

(3) 4.98 cm

(4) 5.00 cm

45) A screw gauge has 50 circular scale divisions (C.S.D.) and advances 0.5 mm along the main scale in one complete rotation. While measuring the diameter of a wire, the main scale reading (M.S.R.) is 3.5 mm and the circular scale reading (C.S.R.) is 32 divisions. If the screw gauge has a positive zero error of 0.06 mm, determine the corrected diameter of the wire.

(1) 3.82 mm

(2) 3.76 mm

(3) 3.88 mm

(4) None of these

CHEMISTRY

1) The number of atoms of Cr and O are 4.8×10^{10} and 9.6×10^{10} respectively. Its empirical formula is –

(1) Cr_2O_3

(2) CrO_2

(3) Cr_2O_4

(4) CrO_5

2) If 0.5 mole of $BaCl_2$ is mixed with 0.2 mole of Na_3PO_4 the maximum number of moles of $Ba_3(PO_4)_2$ that can be formed is -

$3BaCl_2 + 2Na_3PO_4 \rightarrow Ba_3(PO_4)_2 + 6NaCl$

(1) 0.7
(2) 0.5
(3) 0.3
(4) 0.1

3) The quantum number of 20th electron of Fe(Z = 26) would be :–

(1) $3, 2, -2, -\frac{1}{2}$
(2) $3, 2, 0, \frac{1}{2}$
(3) $4, 0, 0, +\frac{1}{2}$
(4) $4, 1, -1, +\frac{1}{2}$

4) Wavelength of first line of Lyman series in terms of R of hydrogen atom :-

(1) $\frac{3R}{4}$
(2) $\frac{4}{3R}$
(3) $\frac{5R}{36}$
(4) $\frac{36}{5R}$

5) The equilibrium constant of the following are:
$N_2 + 3H_2 \rightleftharpoons 2NH_3$; K_1
$N_2 + O_2 \rightleftharpoons 2NO$; K_2
$H_2 + \frac{1}{2}O_2 \rightarrow H_2O$; K_3
The equilibrium constant (K) of the reaction :
$2NH_3 + \frac{5}{2}O_2 \overset{K}{\rightleftharpoons} 2NO + 3H_2O$, will be :

(1) $K_1K_3^3/K_2$
(2) $K_2^3K_3/K_1$
(3) K_2K_3/K_1
(4) $K_2K_3^3/K_1$

6) For the equilibrium reaction
$A(g) \rightleftharpoons B(g) + C(g) + \frac{1}{2}D(g)$

If the molar mass of A is 48 and the vapour density at equilibrium mixture is 16, then calculate degree of dissociation of A.

(1) $\frac{1}{3}$

(2) $\frac{5}{3}$

(3) $\frac{4}{3}$

(4) $\frac{2}{3}$

7) If degree of hydrolysis of 0.1 M NH_4CN is found to be 50%. Then degree of hydrolysis of its 0.2M solution will be

(1) 50%

(2) 25%

(3) 20%

(4) 10%

8) Which of the following is a buffer solution ?

(1) 500mL of 0.1 N CH_3COOH + 500 mL of 0.1N HCl

(2) 500 mL of 0.2 N CH_3COOH + 500 mL to 0.1N NaOH

(3) 500 mL of 0.1 N CH_3COOH + 500 mL of 0.2N NaOH

(4) 500mL of 0.1 N CH_3COOH + 500 mL of 0.1N NaOH

9) The pH of a simple sodium acetate buffer is given by pH = $pK_a + \log \frac{[Salt]}{[Acid]}$
K_a of acetic acid = 1.8×10^{-5}. If [Salt] = [Acid] = 0.1 M, the pH of the solution would be about

(1) 7

(2) 1.4

(3) 4.75

(4) 5.3

10) What is the pH of a saturated solution of $Cu(OH)_2$? ($K_{sp} = 3.2 \times 10^{-17}$)

(1) 6.17

(2) 7.90

(3) 7.30

(4) 8.60

11) When ethyne is passed through a red hot tube, then formation of benzene takes place :
$\Delta H^{o}_{f(C_2H_2)(g)} = 230\ kJ\ mol^{-1}$
$\Delta H^{o}_{f(C_6H_6)(g)} = 85\ kJ\ mol^{-1}$
Calculate the standard heat of trimerisation of ethyne to benzene.

$3C_2H_{2\ (g)} \rightarrow C_6H_{6\ (g)}$

(1) –605 kJ mol^{-1}

(2) –205 kJ mol^{-1}

(3) 205 kJ mol^{-1}

(4) 605 kJ mol^{-1}

12) The enthalpy changes for two reactions are given by the equations:

$2Cr(s) + \frac{3}{2}O_2(g) \rightarrow Cr_2O_3\ (s);$ $\Delta H = -1130$ kJ

$C(s) + \frac{1}{2}O_2(g) \rightarrow CO(g);$ $\Delta H = -110$ kJ

What is the enthalpy change in kJ for the following reaction?

$3C(s) + Cr_2O_3(s) \rightarrow 2Cr(s) + 3CO(g)$

(1) –800 kJ

(2) +800 kJ

(3) +1020 kJ

(4) –1460 kJ

13) Which of the following examples does not represent disproportionation?

(1) $4KClO_3 \rightarrow 3KClO_4 + KC$

(2) $2H_2O_2 \rightarrow 2H_2O + O_2$

(3) $MnO_2 + 4HCl \rightarrow MnCl_2 + Cl_2 + 2H_2O$

(4) $3Cl_2 + 6NaOH \rightarrow 5NaCl + NaClO_3 + 3H_2O$

14) In the following reaction,

$$3Br_2 + 6CO_3^{2-} + 3H_2O \rightarrow 5Br^- + BrO_3^- + 6HCO_3^-$$

(1) bromine is reduced, carbonate is oxidised

(2) bromine is oxidised, carbonate is reduced

(3) bromine is reduced as well as oxidised

(4) bromine is neither reduced nor oxidised

15) The oxidation potentials of Zn, Cu, Ag, H_2 and Ni are 0.76, –0.34, –0.80, 0.00, 0.25 volt respectively. Which of the following reaction will provide maximum voltage?

(1) $Zn + Cu^{2+} \rightarrow Cu + Zn^{2+}$

(2) $Zn + 2Ag^+ \rightarrow 2Ag + Zn^{2+}$

(3) $H_2 + Cu^{2+} \rightarrow 2H^+ + Cu$

(4) $H_2 + Ni^{2+} \rightarrow 2H^+ + Ni$

16) For a spontaneous reaction, ΔG, equilibrium constant (K) and E°_{cell} will be respectively:

(1) –ve, > 1, + ve

(2) +ve, > 1, – ve

(3) –ve, < 1, – ve

(4) –ve, > 1, –ve

17) Consider the following cell reaction :
$2Fe(s) + O_2(g) + 4H^+(aq) \rightarrow 2Fe^{2+}(aq) + 2H_2O(l)$
$E° = 1.67$ V
At $[Fe^{2+}] = 10^{-3}$ M, $P(O_2) = 0.1$ atm and pH =3, the cell potential at 25° C is:

(1) 1.47V

(2) 1.77V

(3) 1.87V

(4) 1.57V

18) Two liquids X and Y form an ideal solution. The mixture has a vapour pressure of 400 mm at 300 K when mixed in the molar ratio of 1 : 1 and a vapour pressure of 350 mm when mixed in the molar ratio of 1 : 2 at the same temperature. The vapour pressure of the two pure liquids X and Y respectively are

(1) 250 mm, 550 mm

(2) 350 mm, 450 mm

(3) 350 mm, 700 mm

(4) 550 mm, 250 mm

19) A and B react as per the reaction
$A(g) + 2B(g) \rightarrow C(g)$
From the data given below identify the Rate Law

[A] (M)	[B] (M)	ROR ($M\ sec^{-1}$)
0.3	0.4	2×10^{-3}
0.6	0.8	8×10^{-3}
0.6	0.4	4×10^{-3}

(1) $R = k [A]^2 [B]^1$

(2) $R = k [A] [B]$

(3) $R = k [A]^1 [B]^0$

(4) $R = k [A] [B]^{-1}$

20) For a first order reaction, the concentration decreases to 30% of its initial value in 5.0 min. What is the rate constant ?

(1) 0.46 min^{-1}

(2) 0.24 min^{-1}

(3) 0.14 min^{-1}

(4) 0.060 min^{-1}

21) Consider the Arrhenius equation given below and mark the **CORRECT** option, $k = Ae^{-E_a/RT}$ /

(1) Rate constant increases exponentially with increasing activation energy and decreasing temperature.

(2) Rate constant remains constant with increasing activation energy and decreasing temperature.

(3) Rate constant increases exponentially with decreasing activation energy and decreasing temperature.

(4) Rate constant increases exponentially with decreasing activation energy and increasing temperature.

22) A first order reaction has a rate constant of $0.0541s^{-1}$ at 25°C, What is the half-life for this reaction?

(1) 18.5 s

(2) 12.8 s

(3) 0.0781 s

(4) 0.0375 s

23) Match the column

	Column-I		**Column-II**
(I)	Van't Hoff factor for dissociation	(a)	More than 1
(II)	Van't Hoff factor for association	(b)	Less than 1
(III)	Van't Hoff factor for association for non electrolyte	(c)	May be greater than 1 or less than 1.
(IV)	Van't Hoff factor for electrolyte	(d)	Equal to 1

Which is correct :-

(1) (I) → (a), (II) → (d), (III) → (c), (IV) → (b)

(2) (I) → (d), (II) → (b), (III) → (a), (IV) → (c)

(3) (I) → (b), (II) → (c), (III) → (a), (IV) → (d)

(4) (I) → (a), (II) → (b), (III) → (d), (IV) → (c)

24) Given below are two statements:
Statement-I : There is no effect on equilibrium constant if inert gas is added to the reaction.
$A(g) + B(g) \rightleftharpoons C(g)$
Statement-II : Equilibrium constant changes only with temperature.
In the light of the above statements, choose the most appropriate answer from the options given belows:

(1) Both statement I and statement II are incorrect.

(2) Statement I is correct but statement II is incorrect

(3) Statement I is incorrect but statement II is correct.

(4) Both statements I and statements II are correct.

25) Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A): ΔH_f° for $H_2(g)$, and $Cl_2(g)$ is zero.
Reason (R): ΔH_f° for every element in its stable and standard state is zero.
In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A).**

(2) **(A)** is correct but **(R)** is not correct.

(3) **(A)** is not correct but **(R)** is correct.

(4) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A).**

26) When the substances SiC, KCl, CH_3OH and C_2H_6 are arranged in order of increasing melting point, what is the correct order ?

(1) SiC, KCl, CH_3OH, C_2H_6

(2) CH_3OH, C_2H_6, SiC, KCl

(3) KCl, SiC, C_2H_6, CH_3OH

(4) C_2H_6, CH_3OH, KCl, SiC

27) In which of the following transformation, the bond order has increased and the magnetic behaviour has changed ?

(1) $C_2^+ \rightarrow C_2$

(2) $NO^+ \rightarrow NO$

(3) $O_2 \rightarrow O_2^+$

(4) $N_2 \rightarrow N_2^+$

28) Assuming pure 2s and 2p orbitals of carbon are used in forming CH_4 molecule, which of the following statement is false ?

(1) Three C–H bonds will be at right angle

(2) One C–H bond will be weaker than other three C–H bonds

(3) The shape of molecule will be tetrahedral

(4) The angle of C–H bond formed by s-s overlapping will be uncertain with respect to other three bonds.

29) Match the Following -

	Column I		**Column II**
(A)	Ionic Bond	(1)	High melting point, Soluble in water
(B)	Non-polar covalent Bond	(2)	Low melting point, Insoluble in water
(C)	Metallic Bond	(3)	High electrical conductivity, Malleable
(D)	Polar Covalent Bond	(4)	Soluble in water, may form hydrogen bonds

(1) A → 1, B → 2, C → 3, D → 4

(2) A → 3, B → 4, C → 1, D → 2

(3) A → 2, B → 4, C → 1, D → 3

(4) A → 4, B → 3, C → 2, D → 1

30) Which of the following orbital combination can not form π-bond ?

(1) $p_x + p_x$ sideways overlapping

(2) $d_{x^2-y^2} + p_y$ sideways overlapping

(3) $d_{xy} + d_{xy}$ sideways overlapping

(4) $d_{yz} + p_y$ sideways overlapping

31) Which of the following compounds has dipole moment approximately equal to that of chlorobenzene ?

(1) o-dichlorobenzene

(2) m-dichlorobenzene

(3) p-dichlorobenzene

(4) p-chloronitrobenzene

32)

The bond angle of NH_3, NH_4^+ and NH_2^- are in the order :

(1) $NH_2^- > NH_3 > NH_4^+$

(2) $NH_4^+ > NH_3 > NH_2^-$

(3) $NH_3 > NH_2^- > NH_4^+$

(4) $NH_3 > NH_4^+ > NH_2^-$

33) The boiling points at atmospheric pressure of HF, H_2S, NH_3 can be arranged in the following order.

(1) HF > NH_3 > H_2S

(2) HF > H_2S > NH_3

(3) HF > H_2S < NH_3

(4) HF < NH_3 < H_2S

34) In terms of polar character, which of the following order is correct ?

(1) $NH_3 < H_2O < HF < H_2S$

(2) $H_2S < NH_3 < H_2O < HF$

(3) $H_2O > NH_3 < H_2S < HF$

(4) $HF < H_2O < NH_3 < H_2S$

35) Given below are two statements.
Statement-1 : N_2 and NO^+ both are diamagnetic substances.

Statement-2: NO^+ is isoelectronic with N_2.
In the light of the above statements, choose the most appropriate answer from the options given below :

(1) Both statement-I and statement-II are correct.
(2) Statement-I is correct but statement-II is incorrect.
(3) Both statement-I and statement-II are incorrect.
(4) Statement-I is incorrect but statement-II is correct.

36) Which of the following has maximum solubility in the below mentioned alkaline earth metal sulphates ?

(1) $BaSO_4$
(2) $CaSO_4$
(3) $SrSO_4$
(4) $BeSO_4$

37) The diagonal partner of element B is :

(1) Li
(2) Al
(3) Si
(4) Mg

38) The element Z = 114 has been discovered recently. It will belong to which of the following family/group and electronic configuration?

(1) Carbon family, [Rn] $5f^{14}\ 6d^{10}\ 7s^2\ 7p^2$
(2) Oxygen family, [Rn] $5f^{14}\ 6d^{10}\ 7s^2\ 7p^4$
(3) Nitrogen family, [Rn] $5f^{14}\ 6d^{10}\ 7s^2\ 7p^6$
(4) Halogen family, [Rn] $5f^{14}\ 6d^{10}\ 7s^2\ 7p^5$

39) The first ionisation enthalpies of Na, Mg, Al and Si are in the order:

(1) Na < Mg > Al < Si
(2) Na > Mg > Al > Si
(3) Na < Mg < Al < Si
(4) Na < Mg < Al > Si

40) Which of the following groups of elements is assigned zero electronegativity :

(1) Alkali metals
(2) Noble gases
(3) Alkaline earth metals
(4) Rare earth metals

41) Which of the following can readily react with NaOH solution :

(1) CO
(2) NO
(3) CaO
(4) Cl_2O_7

42) If the ionic radius of K^+ and F^- are nearly same 1.34 Å, then the atomic radii of K and F respectively will be :

(1) 1.96 Å, 1.34 Å
(2) 0.72 Å, 1.96 Å
(3) 1.96 Å, 0.72 Å
(4) 1.34 Å, 1.34 Å

43) Which of the following can't exist :

(1) AlF_6^{3-}
(2) BF_4^-
(3) BF_6^{3-}
(4) Both 1 and 2

44) The ___ of any given element is not constant, it varies depending on the element to which it is bound :

(1) Ionic size
(2) Ionic charge
(3) Electronegativity
(4) Ionisation energy

45) IUPAC name of the element with atomic number as 117 is :

(1) Ununheptium
(2) Unnilheptium
(3) Unnilseptium
(4) Ununseptium

BIOLOGY

1) Match the following:

Column I (Phylum)	Column II (Special Characters)	Column III (Example)
A. Porifera	1. Canal system	i. Sycon
B. Cnidaria	2. Cnidoblasts	ii. Physalia
C. Annelida	3. Metamerism	iii. Nereis

(1) A–1–i, B–2–ii, C–3–iii

(2) A–2–i, B–1–ii, C–3–iii

(3) A–1–ii, B–2–i, C–3–iii

(4) A–3–i, B–2–ii, C–1–iii

2) Consider the following statements:
1. Adult echinoderms are radially symmetrical.
2. Nematoda are pseudocoelomates.
3. Annelida possess a closed circulatory system.
4. Arthropoda are triploblastic but acoelomate.
5. Mollusca have an open circulatory system.

How many statements are correct?

(1) 2

(2) 3

(3) 4

(4) 5

3) Match the classes of Arthropoda with their correct characters:

	Class		Character
A.	Insecta	1.	Two pairs of antennae
B.	Arachnida	2.	Three pairs of legs
C.	Crustacea	3.	No antennae

(1) A–2, B–3, C–1

(2) A–1, B–2, C–3

(3) A–3, B–2, C–1

(4) A–2, B–1, C–3

4) Which of the following are correct examples of Cnidaria?
A. Hydra
B. Aurelia
C. Adamsia
D. Pleurobrachia

(1) A and B only

(2) A, B, C only

(3) B, C, D only

(4) A, C, D only

5) **Assertion (A):** Amphibians can live both on land and in water.
Reason (R): They possess moist skin for cutaneous respiration.

(1) Both A and R are true and R is correct explanation of A.

(2) Both A and R are true but R is not correct explanation of A.

(3) A is true but R is false.

(4) A is false but R is true.

6) Match the following:

	Column I		Column II
A.	Ampulla	1.	Implantation
B.	Isthmus	2.	Narrow part of oviduct
C.	Endometrium	3.	Site of fertilisation
D.	Cervix	4.	Lower part of uterus

(1) A–2, B–3, C–1, D–4

(2) A–3, B–1, C–2, D–4

(3) A–3, B–2, C–1, D–4

(4) A–1, B–2, C–3, D–4

7) Consider the following statements:
1. Spermatogenesis begins at puberty.
2. LH stimulates Leydig cells to secrete testosterone.
3. FSH acts on Sertoli cells.
4. Ovulation occurs during luteal phase.
5. Corpus luteum secretes progesterone.
How many statements are correct?

(1) 2

(2) 3

(3) 4

(4) 5

8) Identify the correct sequence of events:
A. Cleavage
B. Fertilisation
C. Implantation
D. Gastrulation

(1) A → B → C → D

(2) B → C → A → D

(3) B → A → C → D

(4) B → A → D → C

9) **Assertion (A):** Human placenta acts as an endocrine tissue.
Reason (R): It secretes hCG, hPL, estrogens and progesterone.

(1) Both A and R are true and R is correct explanation of A

(2) Both A and R are true but R is not correct explanation of A

(3) A is true but R is false

(4) A is false but R is true

10) **Statement A:** Zona pellucida blocks polyspermy.
Statement B: It ensures only one sperm fertilises the ovum.

(1) Both A and B are correct and B explains A.

(2) Both A and B are correct but B does not explain A.

(3) A is correct but B is incorrect.

(4) A is incorrect but B is correct.

11) Consider the following statements: 1. Menstrual phase involves breakdown of endometrial lining.
2. Follicular phase is also called proliferative phase.
3. Ovulation occurs on 14th day of 28-day cycle.
4. Progesterone level is highest during follicular phase.
How many statements are correct?

(1) 1

(2) 2

(3) 3

(4) 4

12) Consider the following statements:
1. Reproductive health refers to total well-being in all aspects of reproduction.
2. India initiated family planning programmes in 1951.
3. Lactational amenorrhea is a natural method of contraception.
4. IUDs increase phagocytosis of sperms in uterus.
5. Vasectomy involves removal of ovaries.
How many statements are correct?

(1) 2

(2) 3

(3) 4

(4) 5

13) **Assertion (A):** Oral contraceptive pills are widely used by females.
Reason (R): They contain hormones that inhibit ovulation.

(1) Both A and R are true and R is correct explanation of A

(2) Both A and R are true but R is not correct explanation of A

(3) A is true but R is false

(4) A is false but R is true

14) **Statement A:** Amniocentesis is used for detecting fetal abnormalities.
Statement B: Its misuse has led to female foeticide.

(1) Both A and B are correct and B explains A.

(2) Both A and B are correct but B does not explain A.

(3) A is correct but B is incorrect.

(4) A is incorrect but B is correct.

15) Match the following:

	Column I (Tissue)		Column II (Function)
A.	Squamous epithelium	1.	Secretion
B.	Cuboidal epithelium	2.	Diffusion
C.	Columnar epithelium	3.	Absorption
D.	Cardiac muscle	4.	Involuntary contraction

(1) A–2, B–1, C–3, D–4

(2) A–1, B–2, C–3, D–4

(3) A–2, B–3, C–1, D–4

(4) A–3, B–1, C–2, D–4

16) Consider the following statements regarding frog :
1. Frogs are poikilothermic animals.
2. Their heart is three-chambered.
3. They respire only through lungs.
4. Fertilisation is external.
5. They exhibit metamorphosis.
How many statements are correct?

(1) 2

(2) 3

(3) 4

(4) 5

17) Identify the correct sequence of alimentary canal in cockroach:
A. Mouth
B. Gizzard
C. Crop
D. Midgut

(1) A → B → C → D

(2) A → C → D → B

(3) A → C → B → D

(4) A → B → D → C

18) **Assertion (A):** Cardiac muscles are involuntary.
Reason (R): They show rhythmic contraction and relaxation.

(1) Both A and R are true and R is correct explanation of A.

(2) Both A and R are true but R is not correct explanation of A.

(3) A is true but R is false.

(4) A is false but R is true.

19) **Statement A:** Frogs have a three-chambered heart.
Statement B: Mixing of oxygenated and deoxygenated blood occurs.

(1) Both A and B are correct.

(2) Both A and B are incorrect.

(3) A is correct but B is incorrect

(4) A is incorrect but B is correct

20) Consider the following statements:
1. Cockroach has open circulatory system.
2. Malpighian tubules help in excretion.
3. Striated muscles are involuntary.
4. Neurons transmit impulses.
How many statements are correct?

(1) 1

(2) 2

(3) 3

(4) 4

21) Match the following:

	Column I		Column II
A.	Alveoli	1.	Exchange of gases
B.	Diaphragm	2.	Increases thoracic volume
C.	Haemoglobin	3.	Transport of oxygen
D.	Pleura	4.	Double layered membrane of lungs

(1) A–2, B–1, C–3, D–4

(2) A–1, B–2, C–3, D–4

(3) A–1, B–3, C–2, D–4

(4) A–4, B–2, C–3, D–1

22) Consider the following statements:
1. Inspiration is an active process.
2. Expiration is always active.
3. Oxygen is transported mainly as oxyhaemoglobin.
4. Carbon dioxide is transported mainly as bicarbonate ions.
5. Alveoli provide large surface area.
How many statements are correct?

(1) 2

(2) 3

(3) 4

(4) 5

23) In the alveolus, the site of diffusion of gases is between:
A. Alveolar epithelium and capillary endothelium
B. Bronchi and bronchioles
C. Trachea and bronchi
D. Pleura and lungs

(1) B only

(2) C only

(3) A only

(4) D only

24) Identify the correct pathway of air:
A. Trachea
B. Bronchi
C. Bronchioles
D. Alveoli

(1) A → B → C → D

(2) B → A → C → D

(3) A → C → B → D

(4) A → B → D → C

25) **Assertion (A):** CO_2 diffuses more rapidly than O_2.
Reason (R): CO_2 is more soluble in blood than O_2.

(1) Both A and R are true and R is correct explanation of A

(2) Both A and R are true but R is not correct explanation of A

(3) A is true but R is false

(4) A is false but R is true

26) **Statement A:** Binding of oxygen with haemoglobin is mainly related to partial pressure of oxygen.
Statement B: Higher pO_2 favours formation of oxyhaemoglobin.

(1) Both A and B are correct and B explains A.

(2) Both A and B are correct but B does not explain A.

(3) A is correct but B is incorrect.

(4) A is incorrect but B is correct.

27) Consider the following statements:
1. Tidal volume is about 500 mL.
2. Vital capacity includes IRV, TV and ERV.
3. Residual volume can be measured by spirometer.
4. Total lung capacity includes residual volume.
How many statements are correct?

(1) 1

(2) 2

(3) 3

(4) 4

28) Match the following:

	Column I		**Column II**

A.	Plasma	1.	Clotting
B.	RBC	2.	Transport of gases
C.	WBC	3.	Defence
D.	Platelets	4.	Fluid matrix

(1) A–2, B–4, C–3, D–1

(2) A–4, B–3, C–2, D–1

(3) A–1, B–2, C–3, D–4

(4) A–4, B–2, C–3, D–1

29) Consider the following statements:
1. RBCs lack nucleus in mammals.
2. Haemoglobin is present in WBCs.
3. Platelets help in clotting.
4. Plasma contains proteins like albumin and fibrinogen.
5. Lymph contains RBCs.
How many statements are correct?

(1) 2

(2) 3

(3) 4

(4) 5

30) Identify the correct sequence of cardiac cycle events:
A. Ventricular systole
B. Atrial systole
C. Joint diastole

(1) A → B → C

(2) B → A → C

(3) C → B → A

(4) B → C → A

31) **Assertion (A):** SA node is called pacemaker of the heart.
Reason (R): It initiates the cardiac impulse.

(1) Both A and R are true and R is correct explanation of A.

(2) Both A and R are true but R is not correct explanation of A.

(3) A is true but R is false.

(4) A is false but R is true.

32) **Statement A:** Double circulation is present in humans.
Statement B: Blood passes through heart twice in one complete cycle.

(1) Both A and B are correct and B explains A

(2) Both A and B are correct but B does not explain A

(3) A is correct but B is incorrect

(4) A is incorrect but B is correct

33) Consider the following statements:
1. Arteries carry blood away from the heart.
2. Veins have thick muscular walls than arteries.
3. Capillaries allow exchange of materials.
4. Hepatic portal system carries blood from intestine to liver.
How many statements are correct?

(1) 1

(2) 2

(3) 3

(4) 4

34) Match Column I with Column II regarding sex determination mechanisms.

	Column I		**Column II**
A.	Grasshopper	P.	Female heterogamety
B.	Birds	Q.	XX–XO type
C.	Honey bee	R.	Haplodiploidy
D.	Human beings	S.	XX–XY type

(1) A–Q, B–P, C–R, D–S

(2) A–S, B–Q, C–R, D–P

(3) A–Q, B–R, C–P, D–S

(4) A–P, B–Q, C–S, D–R

35) In a dihybrid cross AaBb × AaBb, what fraction of progeny will be heterozygous for gene A and homozygous recessive for gene B?

(1) 1/4

(2) 3/8

(3) 1/8

(4) 1/2

36) A male individual shows gynaecomastia and has overall masculine development. Cytological examination reveals 47 chromosomes with an extra X chromosome. The individual is suffering from

(1) Turner’s syndrome

(2) Down’s syndrome

(3) Klinefelter’s syndrome

(4) Sickle-cell anaemia

37) A pedigree chart shows that all affected individuals are males and the trait is transmitted from affected father to all sons.
The trait is most likely:

(1) Autosomal dominant

(2) X-linked recessive

(3) Y-linked

(4) Autosomal recessive

38) **Assertion (A):** The diameter of the DNA double helix remains constant throughout its entire length.
Reason (R): AT base pair has double hydrogen bonds while **GC** base pair has triple hydrogen bonds.

(1) Both A and R are true, and R is the correct explanation of A.

(2) Both A and R are true, but R is not the correct explanation of A.

(3) A is true, but R is false.

(4) A is false, but R is true.

39) During an experiment, a student observed that removal of phosphate groups from nucleotides prevented formation of a polynucleotide chain. This is because phosphate groups are involved in:

(1) Hydrogen bond formation

(2) Glycosidic linkage formation

(3) Phosphodiester bond formation

(4) Complementary base pairing

40) Who experimentally proved semiconservative replication in chromosomes?

(1) Meselson and Stahl

(2) Taylor

(3) Hershey and Chase

(4) Watson and Crick

41) Select the correct statements-
A. The VNTR belongs to a class of satellite DNA referred to as mini-satellite.
B. DNA from a single cell is not enough to perform DNA fingerprinting analysis.
C. DNA fingerprinting is used in determining population and genetic diversities.
D. Sanger is credited for developing method for determination of amino acid sequences in proteins.

(1) A, B, C and D

(2) A, B and D

(3) A, C and D

(4) B and D

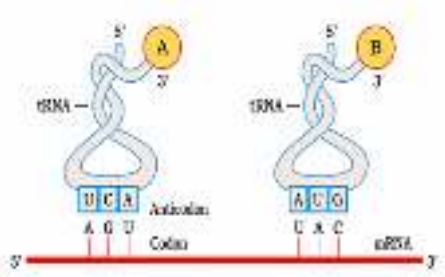
42) Incorrect statement about mutation is :

(1) Mutation may result in loss or gain of a gene and so a function.

(2) Effects of large deletions and rearrangements in a segment of DNA are difficult to comprehend.

(3) When a standard codon is removed or deleted, it can not be called frameshift mutation.

(4) The reason a frameshift mutation is so destructive is precisely because the genetic code is commaless.



43) Identify the amino acids carried by these tRNAs respectively.

(1) Tyr, Ser
(2) Ser, Tyr
(3) Ala, Val
(4) Val, Ala

44) A pure tall pea plant is crossed with dwarf. This cross produces 400 offsprings. How many of them are heterozygous tall ?

(1) 200
(2) 100
(3) 400
(4) 300

45) What results Mendel would have got when he self-pollinated an impure tall F_2 garden pea plant ?

(1) All tall plants
(2) All drwarf plants
(3) Tall and dwarf plants in 3 : 1 ratio
(4) Tall and dwarf plants in 1 : 1 ratio

46) Whorl of carpels in flower represents?

(1) Gynoecium
(2) Androecium
(3) Calyx
(4) Corolla

47) Given below are two statements: One is labelled as Assertion (A) and the other is labelled as Reason (R):
Assertion (A) : The generative cell is small and floats in the cytoplasm of the vegetative cell.
Reason (R) : Cells of the nucellus have abundant reserve food materials.
In the light of the above statements, choose the correct answer from the options given below:

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
(2) Both Assertion and Reason are true but Reason is NOT the correct explanation of Assertion.
(3) Assertion is true but Reason is false.
(4) Assertion is false but Reason is true.

48) Find out the incorrect statement :
A. Pollination does not guarantee the transfer of the right type of pollen.
B. All the events from pollen deposition on stigma until double fertilisation are together referred to as pollen-pistil interaction.
C. Chemical interaction between pollen & pistil is a dynamic process.
D. Pistil has the ability to recognise the right type pollen and wrong type pollen.

(1) A, B and C only
(2) A and C Only
(3) C and D Only
(4) Only B

49) Read all statement carefully and select correct statement :

(1) In Gymnospermic plant, formation of endosperms occurs after fertilization.
(2) Coleorhiza and coleoptile are generally present in dicot embryo.
(3) In Grasses family cotyledon is called perisperm
(4) Castor is a dicot and it has endosperm.

50) Find the correct statements from the following and give answer :-
(A) Family is a group of related genera with more number of similarities than genus and species.
(B) Every taxonomic category / Rank / Taxon represents a unit of classification.

(1) Both (A) and (B)
(2) only (B)
(3) only (A)
(4) None

51) **Assertion :** Classification systems have undergone several changes over a period of time.
Reason : This happened, because the criteria for classification get changed gradually.

(1) Both Assertion and Reason are true, and Reason is the correct explanation of Assertion.
(2) Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.
(3) Assertion is true, but Reason is false.
(4) Assertion is false, but Reason is true.

52) Find the correct statements from the following :-
(a) Biology is the science of life forms and living processes.
(b) Biology is the story of life on earth.
(c) Biology is the story of evolution of living organisms on earth.

(1) Only (a) and (b)
(2) Only (c)
(3) Only (a) & (c)
(4) All (a), (b) & (c)

53) **Assertion :** All living organisms present, past and future are linked to one another.

Reason : All living organisms share the common genetic material but to varying degrees.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

54) Consider the following statements :
(a) In two kingdom classification bacteria, blue green algae, fungi, mosses, ferns, gymnosperm and angiosperms are included in plantae kingdom.
(b) In three domain system the eukaryotic organisms are divided into two domains.
(c) According to five kingdom system *Chlamydomonas* and *Chlorella* were placed in plantae kingdom.
Choose the correct answer from the option given below :

(1) (a), (b) and (c) all are correct.

(2) Only (c) is correct.

(3) Only (b) is correct.

(4) Only (a) is correct.

55) **Assertion (A) :** Members of phycomycetes are saprophytic, decomposers, parasitic and found on aquatic habitat.
Reason (R) : The mycelium of members of phycomycetes is septate and coenocytic.

(1) Both (A) and (R) are correct and (R) is the correct explanation of (A).

(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A).

(3) (A) is correct but (R) is incorrect.

(4) Both (A) and (R) are incorrect.

56) Which one of the following option give the correct categorisation of six organisms according to their related kingdoms (A, B and C) ?

	(A) Monera	**(B) Protista**	**(C) Plantae**
(1)	Mycoplasma, Archaebacteria	Protozoa, Dinoflagellates	Mango, Wheat
(2)	Fungi, Mycoplasma	Porifera, Protozoa	Bacteria, Mango
(3)	Eubacteria, Protozoa	Mycoplasma, Diatoms	Wheat, Potato
(4)	Eubacteria, Archaebacteria	Mycoplasma, Wheat	Protozoa, Mango

(1) 1

(2) 2

(3) 3

(4) 4

57) Identify the correct match from column I, II and III.

Column I (Kingdom)		Column II (Cell wall)		Column III (Body organisation)	
1	Monera	a	Present in some	i	Tissue/ organ
2	Protista	b	Present (Chitin)	ii	Loose tissue
3	Fungi	c	Present (Cellulose)	iii	Cellular
4	Plantae	d	Polysaccharide + Amino acid	iv	Cellular

(1) 2–a–iii, 1–d–iv, 3–b–ii, 4–c–i

(2) 1–d–iv, 2–b–i, 3–a–ii, 4–c–iii

(3) 1–d–iv, 2–c–ii, 3–a–i, 4–b–iii

(4) 2–b–iii, 1–c–iv, 3–d–ii, 4–c–i

58) Consider the following statements:-
(a) They grow in soil, on logs and tree stumps and in living plant bodies as parasites. (b) The mycelium is branched and septate and the asexual spores are generally not found but vegetative reproduction by fragmentation is common. (c) The sex organs are absent but plasmogamy is brought about by fusion of two vegetative or somatic cells of different strains or genotypes. Above statements are related to the member of which class :-

(1) Phycomycetes

(2) Ascomycetes

(3) Basidiomycetes

(4) Deuteromycetes

59) **Assertion:** Viruses are non cellular organism.
Reason: They do not perform cellular function like metabolism and growth when present out side host body.

(1) Both Assertion and Reason are true and the Reason is a correct explanation of the Assertion.

(2) Both Assertion and Reason are true and the Reason is not a correct explanation of the Assertion.

(3) Assertion is true but Reason is false.

(4) Both assertion and Reason are false.

60)

Which of the following statements are correct?
(a) Asexual reproduction is absent in Deuteromycetes
(b) Plasmogamy is brought about by fusion of two vegetative cells of same strain only in Basidiomycetes
(c) Basidiospores and ascospores are produced exogenously and endogenously respectively
(d) Bracket fungi is another name of some basidiomycetes

(1) a, b

(2) a, c

(3) c, d

(4) b, c

61) **Statement-1 :** Yeasts are unicellular fungi.
Statement-2 : Dinoflagellates have two flagella one is transverse and other is longitudinal.
Choose the correct option from the given below :-

(1) Both statement-1 and statement-2 are incorrect.

(2) Statement-1 is incorrect but statement-2 is correct.

(3) Statement-1 is correct but statement-2 is incorrect.

(4) Both statement-1 and statement-2 are correct.

62) Read following statements regarding members of phaeophyceae.
(A) Food is stored as complex proteins which may be in the form of laminarin or mannitol.
(B) They possess algin in addition to cellulose in their cell walls.
(C) Zoospores are pear-shaped and have two equal laterally attached flagella.

(1) Only (A) is correct

(2) Only (A) is incorrect

(3) Only (C) is incorrect

(4) Both (A) and (C) are incorrect.

63) **Assertion (A) :** Members of rhodophyceae occur in both well-lighted regions and at great depths in oceans where relatively little light penetrates.
Reason (R) : They have the predominance of the red pigment, r-phycocyanin in their body.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true and (R) is not the correct explanation of of (A).

(3) Both (A) and (R) are false.

(4) Only (A) is true.

64) **Assertion (A) :** Bryophytes are also called amphibians of the plant kingdom.
Reason (R) : They play an important role in plant succession on bare rocks/soil.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true and (R) is not the correct explanation of (A).

(3) Both (A) and (R) are false.

(4) Only (A) is true.

65) Read the following five statements (A to E) and select the option with **all correct** statements :-
(A) Mosses along with lichens are the first organisms to colonise a rock.
(B) *Selaginella* is a homosporous pteridophyte
(C) Coralloid roots in *Cycas* have mycorhiza.
(D) Main plant body of bryophytes is gametophytic, whereas in pteridophytes it is saprophyte
(E) In gymnosperms, male and female gametophytes are present within sporangia located on sporophyte

(1) (B), (C) and (D)

(2) (A), (D) and (E)

(3) (A) and (E)

(4) (A), (C) and (D)

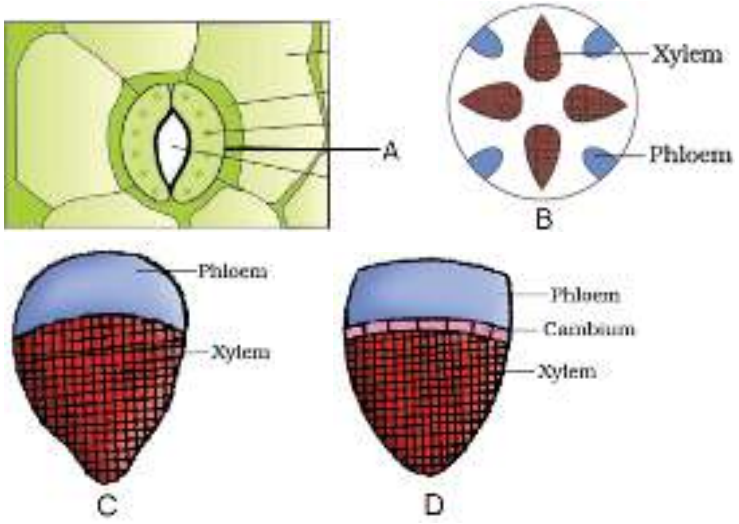
66) Which of the following option have example of primary meristem ?

(1) Apical meristem, Intrafascicular cambium
(2) Intercalary meristem, cork cambium
(3) Intercalary meristem, interfascicular cambium
(4) Intra fascicular cambium, interfascicular cambium

67) Consider the following statements and select the incorrect statement :-

(1) Axillary bud is derived from shoot apical meristem.
(2) In sunflower stem, sclerenchymatous pericycle is located just above phloem bundle in semi lunar patches.
(3) In conjoint vascular bundle phloem usually located on the inner side of xylem.
(4) Phloem parenchyma is usually not present in monocot stem.

68) Identify the (A), (B), (C) and (D) in the given figures and select the correct option :-



	(A)	(B)	(C)	(D)
(1)	Guard cells	Conjoint closed	Conjoint open	Radial condition
(2)	Subsidiary cells	Conjoint closed	Conjoint open	Conjoint condition
(3)	Guard cells	Radial condition	Conjoint closed	Conjoint Bicollateral
(4)	Guard cells	Radial condition	Conjoint closed	Conjoint open

(1) 1
(2) 2
(3) 3
(4) 4

69) Drupe develops from __**A**__ superior ovaries and are __**B**__ seeded. **A** and **B** are :

(1) Monocarpellary, Multi seeded

(2) Multicarpellary, One seeded

(3) Bicarpellary, One seeded

(4) Monocarpellary, One seeded

70) Which of the following is correct regarding seeds in angiosperms :
(A) The ovules after fertilization develops into seeds.
(B) A seed is made up of seed coat and an embryo.
(C) The embryo is made up of a radicle, an embryonal axis and one cotyledon (as in gram or pea) or two cotyledons (as in wheat and maize).
(D) The outer most covering of seed is known as seed coat.
Choose the most appropriate answer from the options given below.

(1) A and B

(2) A, B and C

(3) A, B and D

(4) All A, B, C and D

71) Given below are two statements :
Statement-I : In plants such as bean, gram and pea, the endosperm is not present in mature seeds and such seeds are called non endospermous.
Statement-II : The cotyledons are often fleshy and full of reserve food materials in dicots.
In the light of the above statements, choose the correct answer from the options given below :

(1) Both statement I and statement II are true.

(2) Statement I is true but statement II is false.

(3) Statement I is false but statement II is true.

(4) Both statement I and statement II are false.

72) (1) In monocot the plumule and radicle of seed are enclosed in sheaths which are known as __**A**__ and __**B**__ respectively.
(2) __**C**__ the hilum is a small pore called the micropyle.
A, B and C are respectively ?

(1) A–Coleoptile, B–Coleorhiza, C–Above

(2) A–Coleoptile, B–Coleorhiza, C–Below

(3) A–Coleorhiza, B–Coleoptile, C–Above

(4) A–Coleorhiza, B–Coleoptile, C–Below

73) Given below are two statements :
Statement-I : The fruit is characteristic feature of the flowering plants. It is a mature ripened ovary, developed after fertilization.
Statement-II : Micropyle is a scar on the seed coat through which the developing seeds were attached to the fruit.
In the light of the above statements, choose the correct answer from the options given below :

(1) Both statement I and statement II are incorrect.

(2) Statement I is correct but statement II is incorrect.

(3) Statement I is incorrect but statement II is correct.

(4) Both statement I and statement II are correct.

74) How many statements are correct with reference to cruciferae family.
(A) Flower is hypogynous.
(B) Androecium is tetradynamous.
(C) Flower is tetramerous.
(D) Flowers are mainly zygomorphic.

(1) Four

(2) Three

(3) Two

(4) One

75) Which of the following is know as traffic police in cell ?

(1) Endoplasmic Reticulam

(2) Chloroplast

(3) Mitochondria

(4) Golgi Bodies

76) Given below are two statements :
Statement - (I) : ATP synthase is responsible for the synthesis of ATP.
Statement - (II) : Chloroplasts are responsible for the cellular respiration of cell.
Choose the correct option from the given below :

(1) Statement I is correct and statement II is incorrect.

(2) Statement II is incorrect and statement I is correct.

(3) Statement I and statement II both are correct.

(4) Statement I and statement II both are incorrect.

77) **Statement-I :** Schleiden and Schwann together formulated the cell theory.
Statement-II : Rudulf Virchow give the cell theory a final shape.

(1) Both Statement-I and Statement-II are incorrect.

(2) Both Statement-I and Statement-II are correct.

(3) Statement-I is correct and Statement-II is incorrect.

(4) Statement-I is incorrect and Statement-II is correct.

78) **Assertion :** *Mycoplasma,* are the smallest cells.
Reason : Bacteria length could be 3 to 5 μm.

(1) Assertion and Reason both are correct and Reason is correct explanation of Assertion.

(2) Assertion and Reason both are correct but Reason is not correct explanation of Assertion.

(3) Assertion is incorrect and Reason is correct.

(4) Assertion and Reason are incorrect.

79) Which of the following statements is correct ?

(1) The cytoskeleton, made up of microtubules, microfilaments and intermediate filaments, does not play a role in maintaining the shape of the cell.

(2) Cilia and flagella, which are hair-like outgrowths of the cell membrane, do not emerge from a centriole-like structure called the basal bodies.

(3) In a centrosome, the centrioles lie parallel to each other and are made up of nine evenly spaced peripheral fibrils of tubulin protein.

(4) The central part of the proximal region of the centriole, also known as the hub, is connected with tubules of the peripheral triplets by radial spokes made of protein.

80) Match the following columns :-

	Column-I		**Column-II**
I.	Pigments	a.	Vinblastin, curcumin
II.	Alkaloids	b.	Abrin, ricin
III.	Toxins	c.	Carotenoids, anthocyanins
IV.	Drugs	d.	Morphine, codeine

(1) I-a, II-b, III-c, IV-d

(2) I-c, II-b, III-a, IV-d

(3) I-c, II-d, III-b, IV-a

(4) I-c, II-a, III-b, IV-d

81)

Find out the correct match between column I and column II.

Column-I		**Column-II**	
(a)	GLUT – 4	**(i)**	Most abundant protein in the biosphere
(b)	Collagen	**(ii)**	Transporter protein
(c)	Insulin	**(iii)**	Enzyme
(d)	Trypsin	**(iv)**	Intercellular ground substance
(e)	RuBisCO	**(v)**	Hormone

(1) a–v, b–iv, c–ii, d–i, e–iii

(2) a–ii, b–iv, c–v, d–iii, e–i

(3) a–iv, b–ii, c–v, d–i, e–iii

(4) a–ii, b–iv, c–iii, d–v, e–i

82) Which of the following are true for polysaccharides?
(i) It is acid insoluble.
(ii) Polysaccharides are long chains of monosaccharides.
(iii) Cellulose is a polymeric polysaccharide consisting of only one type of monosaccharide.
(iv) Starch is a polymeric polysaccharide present as a store house of energy in plant tissues.
(v) Cellulose is a heteropolymer and not a homopolymer.
(vi) Animals and plants have polymeric polysaccharide called glycogen.

(1) (i), (ii), (iii), (iv) and (v)

(2) (i), (ii), (iii) and (iv)

(3) (i), (ii), (iii), (iv) and (vi)

(4) All of these

83) Read the following statement (A-D) :-
(a) Each protein is a polymer of amino acids
(b) Protein is a hetero polymer.
(c) Dietary proteins are the source of essential amino acid
(d) Collagen is the most abundant protein in whole of the biosphere
How many of the above statement are right ?

(1) Three

(2) One

(3) Two

(4) Four

84) Identify the correct statements :
A. The protein part of holoenzyme is called apoenzyme.
B. Prosthetic groups are organic compounds and distinguished from other cofactors in that they are tightly bound to the apoenzyme.
C. The non protein part of holoenzyme is called cofactor.
D. Enzymes are composed of only one polypeptide chain.

(1) A and D only

(2) A, B and D only

(3) B, C only

(4) A, B and C only

85)  Identify the above stage with their respective event :-

(1)	Anaphase II	splitting of the centromere
(2)	Anaphase I	Homologous chromosome separate
(3)	Anaphase I	Splitting of the centromere
(4)	Metaphase I	Bivalent chromosomes align on the equatorial plate

(1) 1

(2) 2

(3) 3

(4) 4

86)

Which of the following is/are incorrect with respect to reduction division?
(A) Diakinesis is marked by terminalization of chiasmata.
(B) Splitting of centromere occurs at anaphase-I.
(C) Pairing of homologous chromosomes occurs in prophase-I.

(D) Interkinesis is generally short lived phase which is followed by prophase-II, a much simpler phase than prophase-I.

(1) A and B

(2) B and D

(3) A and C

(4) Only B

87) A bivalent of meiosis I consists of -

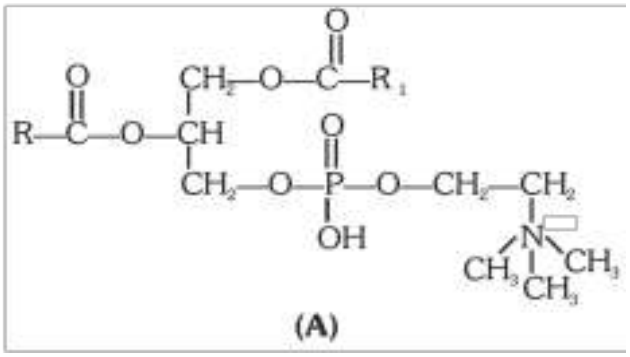
(1) Two chromatids and one centromere

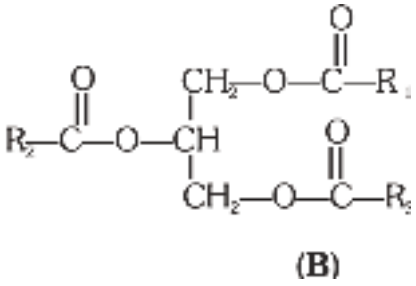
(2) Two chromatids and two centromeres

(3) Four chromatids and two centromeres

(4) Four chromatids and four centromeres

88) The figure shows structure of two lipids A and B. Identify these –





(1) A–triglyceride, B–phospholipid

(2) A–lecithin, B–triglyceride

(3) A–cholesterol, B–glycerol

(4) A–diglyceride, B–monoglyceride

89) Enzymes which catalyze hydrolysis of ester, ether, peptide, glycosidic bond belongs to class :-

(1) Oxidoreductases

(2) Tranferases

(3) Hydrolases

(4) Ligases

90) How many of the given sugars are reducing in nature :-
Glucose, Fructose, Galactose, Maltose, Sucrose, Trehalose, Lactose

(1) 2
(2) 3
(3) 5
(4) 7

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	1	1	3	3	2	1	1	1	3	3	4	2	1	3	2	2	3	3	2	1
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	1	3	2	1	2	1	4	1	2	2	1	2	3	3	3	2	1	2	2	2
Q.	41	42	43	44	45															
A.	1	3	3	3	2															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	2	4	3	2	4	1	1	2	3	4	1	2	3	3	2	1	4	4	2	2
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	4	2	4	4	4	4	1	3	1	2	2	2	1	2	1	4	3	1	1	2
Q.	86	87	88	89	90															
A.	4	3	3	3	4															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	1	3	1	2	1	3	3	3	1	1	3	3	1	2	1	3	3	2	1	3
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	2	3	3	1	1	1	3	4	2	2	1	1	3	1	3	3	3	2	3	2
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	3	2	2	3	3	1	2	4	4	2	1	4	1	4	3	1	1	3	1	3
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	4	4	4	2	3	1	3	4	4	3	1	1	2	2	4	1	2	2	4	3
Q.	171	172	173	174	175	176	177	178	179	180										
A.	2	2	1	4	2	4	3	2	3	3										

SOLUTIONS

PHYSICS

1) **Explaining:** Find electric flux
Concept: Electric flux
Solution:

$$\phi_{net} = \frac{q_{in}}{\varepsilon_0} = \frac{4\pi r^2 \sigma}{\varepsilon_0} = \frac{4\pi 5^2}{\varepsilon_0} \times \frac{\varepsilon_0}{4\pi}$$

ϕ = 25 Nm^2 /C
Final Answer: (1)

2) **Explaining:** Find potential difference $V_A - V_B$
Concept: Apply KVL
Solution:

$$V_A - 6 - 2 \times 3 + \frac{9 \times 10^{-6}}{1 \times 10^{-6}} - 3 \times 3 = V_B$$

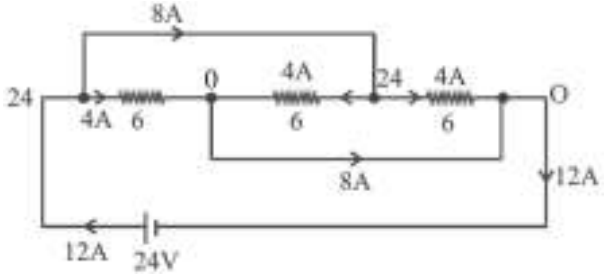
$V_A - 6 - 6 + 9 - 9 = V_B$
$V_A - V_B = 12$ Volt
Final Answer: (1)

3) **Explaining:** Find out amount of charge in capacitor
Concept: Ohm's Law
Solution:

$$I = \frac{V}{R_{net}} = \frac{6}{2.5 + 0.5} = 2A$$

$V_{2\Omega} = IR = 1 \times 2 = 2$ V
$V_{5\mu F} = 2$ V
$Q = CV = 5 \times 2 = 10$ μC
Final Answer: (3)

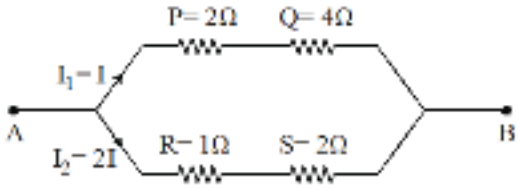
4) **Explaining:** Find incorrect option
Concept: Ohm's law
Solution:



$R_{eq} = 2\Omega$
[All 3 resistances are in parallel]
So total current = $\frac{24}{2}$ = 12 Amp.
Final Answer: (3)

5) **Question Explanation:** We have to calculate the greatest amount of heat produced when current flows from A to B.

Concept: Electrical power and KCL.



$$\frac{I_2}{I_1} = \frac{2}{1}$$

Solution:

$(\text{Power})_P = 2I^2$

$(\text{Power})_Q = 4I^2$

$(\text{Power})_R = 1(2I)^2 = 4I^2$

$(\text{Power})_S = 2(2I)^2 = 8I^2$

Final Answer: (2)

6) **Explaining :** To compare the deflection of charge particle

Concept : This question is based on motion of charge in magnetic field.

Solution :

$$r = \frac{\sqrt{2mK}}{Bq}$$

$H^+ : He^+ : O^{++}$

$$\frac{\sqrt{m}}{q} : \frac{\sqrt{4m}}{q} : \frac{\sqrt{16m}}{2q}$$

1 : 2 : 2

Final Answer : (1)

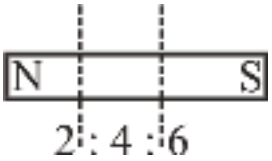
7) **Explaining :** To determine magnetic moment of system.

Concept : This question is based on magnetic moment of bar magnet

Solution :

$$\vec{M}_{net} = \frac{2\mu}{12}\hat{i} + \frac{4\mu}{12}\hat{j} + \frac{6\mu}{12}\hat{k}$$

$$\left|\vec{M}\right| = \sqrt{\frac{7}{18}}\mu$$



Final Answer : (1)

8)

Explaining: Find magnetic field in toroid

Concept: Toroid

Solution:

$$B = \frac{\mu_0 N i}{2\pi R} = \frac{(4\pi \times 10^{-7})(400)(0.9)}{2(9 \times 10^{-2})}\ T$$

$$\left(\text{as } R = \frac{8\text{ cm} + 10\text{ cm}}{2} = 9\text{cm} = 9 \times 10^{-2}\text{m}\right)$$

$= 4\pi \times 10^{-4}$ T

Final Answer: (1)

9) **Explaining :** To predict the path of electron in a uniform magnetic field.

Concept : This question is based on force on moving charge in magnetic field is $\vec{F} = q\left(\vec{v} \times \vec{B}\right)$.

Solution : Clearly, angle between $\vec{v}$ and $\vec{B}$ is 90°. So, motion is circular.

$$\vec{F} = q(\vec{v} \times \vec{B}) = -1.6 \times 10^{-19}((3.2\hat{i} + \hat{j}) \times 2\hat{k})$$

$$= \left(1.6 \times 10^{-19}\right)\left[-2\hat{i} + 6.4\hat{j}\right]$$

Final Answer : (3)

10) **Explaining:** Find ratio of $\frac{i_1}{i_2}$

Concept: Magnetic field due to long wire

Solution:

The magnetic field due to the current-carrying long, straight wire at point a is given by $B = \frac{\mu_0 i}{2\pi a}$. When both the wires carry currents i_1 and i_2 in the same direction, they produce magnetic fields in opposite directions at any point in between the wires.

$$B' = \frac{\mu_0 i_1}{2\pi a} - \frac{\mu_0 i_2}{2\pi a} = 10\mu T \quad \text{..... (1)}$$

Here, a is the distance of the midpoint from both the wires.

When both the wires carry currents in opposite directions, they produce fields in the same direction at the midpoint of the two wires.

$$B = \frac{\mu_0 i_1}{2\pi a} + \frac{\mu_0 i_2}{2\pi a} = 30\mu T \quad \text{..... (2)}$$

On solving eqs. (1) and (2), we get

$$i_1 - i_2 = 10$$
$$i_1 + i_2 = 30$$
$$\Rightarrow i_1 = 20, i_2 = 10$$
$$\Rightarrow \frac{i_1}{i_2} = \frac{2}{1} = 2$$

Final Answer: (3)

11) **Explaining:** Find the ratio of the distances of point X and Y from the centre of the magnet

Concept: Bar magnet

Solution:

$$B_{axis} = B_{eq.} \Rightarrow \frac{\mu_0}{4\pi}\frac{2M}{x^3} = \frac{\mu_0}{4\pi}\frac{M}{y^3}$$

$$\frac{2}{x^3} = \frac{1}{y^3} \Rightarrow \frac{x^3}{y^3} = \frac{2}{1} \Rightarrow \frac{x}{y} = 2^{1/3}$$

Final Answer: (4)

12) **Explaining:** Find the ratio of B/B'
Concept: Magnetic field of square and circular loop
Solution:



$$B = \frac{2\sqrt{2}\mu_0 i}{\pi(1)}$$



$$B' = \frac{\mu_0 i}{2r}$$

$$\frac{B}{B'} = \frac{4\sqrt{2}r}{\pi}$$

$4 = 2\pi r$

$$r = \frac{2}{\pi}$$

$$\Rightarrow \frac{8\sqrt{2}}{\pi^2}$$

Final Answer: (2)

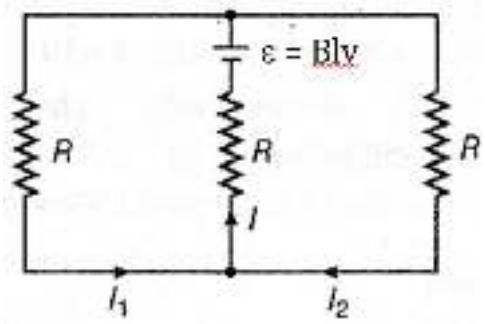
13) **Explaining:** Find the reading of ammeter change if the key k is closed
Concept: Ohm's law
Solution:

$$i_1 = \frac{V}{R_{eq}} = \frac{V}{1.5R} \quad i_2 = \frac{V}{R'_{eq}} = \frac{V}{\frac{4R}{3}}$$

$\Rightarrow i_2 > i_1$

Final Answer: (1)

14) **Explaining :** To find current flowing through resistors.
Concept : This question is based on Motional emf.
Solution : A moving conductor is equivalent to a battery of emf,
$\varepsilon = Blv$



Total resistance

$$= R + \frac{R \times R}{R + R} = \frac{3R}{2}$$

$$\Rightarrow I = \frac{\varepsilon}{3R/2} = \frac{2\varepsilon}{3R} = \frac{2Blv}{3R}$$

$$I_1 = I_2 = \frac{I}{2} = \frac{Blv}{3R}$$

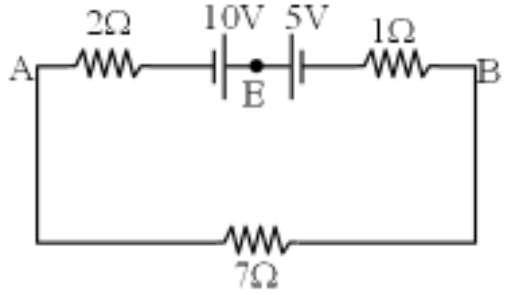
Final Answer : (3)

15) **Explaining:** Identify the correct statement regarding self inductance
Concept: Self inductance
Solution:

A. **Statement A:** The self-inductance of a coil depends on its geometry (area, length, number of turns). This statement is **correct**.
B. **Statement B:** Self-inductance depends on the permeability of the medium. Since the formula for a solenoid's self-inductance is $L = \mu_r \mu_0 n^2 Al$, and μ is the permeability, this statement is **incorrect**.
C. **Statement C:** A self-induced EMF, according to Lenz's law, opposes the change in current. This is the fundamental principle of self-induction. This statement is **correct**.
D. **Statement D:** Self-inductance is a measure of the coil's opposition to changes in current, which is analogous to mass (inertia) opposing changes in motion in mechanics. This statement is **correct**.
E. **Statement E:** To increase the current in an inductor, work must be done against the opposing self-induced EMF. This energy is stored in the magnetic field of the inductor. This statement is **correct**.

The correct statements are A, C, D, and E.
Final Answer: (2)

16) **Explaining :** Given: Single loop circuit with series connected emf source.
Asked: Incorrect statements.
Concept : This question is based on KVl, KCl and ohm's Law.
Solution :
$i = \frac{10-5}{10} = \frac{5}{10} A$ = 0.5 A from A to B through E.



$V_A - V_E = -9V$
$V_E - V_B = 5.5V$
Power supplied by 10V source=10 X 0.5=5W
Final Answer : (2)

17) **Explaining :** To find the value of maximum induced EMF
Concept : This question is based on Periodic EMI
Solution : N = 30 turns
$A = 0.02\ m^2$
$R = 10\ \Omega$
$B = 0.15\ T$
$f = \frac{140}{\pi} Hz$
$e_{max} = ?$

as = $2\pi f = 2\pi \times \frac{140}{\pi} = 280 \text{ rad/sec}$
Now, e = NA Bω sin (ωt)
when sin(ωt) = 1
then $e = e_{max}$
So,
e_{max} = NA Bω (1)
= NABω
= 30 × 0.05 × 0.15 × 280
= 63 volts
Final Answer : (3)

18)

Explaining: Find average EMF
Concept: Flux
Solution:

Initial magnetic flux: $\phi_{initial} = B_{initial} A\cos\theta = (4)(5) = 20 \times \frac{1}{2} = 10$ Wb. Final magnetic flux: $\phi_{final} = B_{final} A \cos\theta = (0)(5)\cos(60°) = 0$ Wb.

Change in flux: $\Delta\phi = \phi_{final} - \phi_{initial} = 0 - 10 = -10$ Wb.

Average induced EMF: $\varepsilon_{avg} = -\frac{\Delta\phi}{\Delta t} = -\frac{-10}{10} = +1V$.
Final Answer: (3)

19) **Explaining:** Choose the correct statement
Concept: Ring
Solution:
(A) V at origin ≠ 0
(B) E = 0
From origin to r = 2, field is towards origin.
(1) $V_{origin} = \frac{kq_1}{r1} + \frac{kq_2}{r2}$

$$V_{origin} = \frac{k(-q)}{\sqrt{6}} + \frac{k\left(2\sqrt{2}q\right)}{4}$$

$V_{origin} \neq 0$
(2) $E_{origin} = 0$

(3) $$E_{x=2} = \frac{kq_1 x}{\left(R_1^2 + x^2\right)^{3/2}} + \frac{kq_2 x}{\left(R_1^2 + x^2\right)^{3/2}}$$

$$E_{x=2} = \frac{k(-q)\,2}{\left[\left(\sqrt{6}\right)^2 + (2)^2\right]^{\frac{3}{2}}} + \frac{k\left(2\sqrt{2}q\right)2}{\left[(4)^2 + (2)^2\right]^{\frac{3}{2}}}$$

$$E_{x=2} = \frac{k(-q)2}{[6+4]^{\frac{3}{2}}} + \frac{k\left(2\sqrt{2}q\right)2}{[16+4]^{\frac{3}{2}}}$$

$$E_{x=2} = \frac{k(-q)2}{10^{\frac{3}{2}}} + \frac{k\left(2\sqrt{2}q\right)2}{20^{\frac{3}{2}}}$$

$$E_{x=2} = \frac{k(-q)2}{10\sqrt{10}} + \frac{k\left(2\sqrt{2}q\right)2}{20\sqrt{20}} = 0$$

Final Answer: (2)

20) **Explaining:** Find EMF of the battery
Concept: Capacitor
Solution:
$q_{7\mu F} = CV = 7\mu \times 6 = 42\ \mu C = q_{3\mu F}$

$$V_{3\mu F} = \frac{Q}{C} = \frac{42\mu}{3} = 14\ \text{Volt}$$

$V_{(7+3)\mu F} = 20$ Volt
Since 3.9 μF are in parallel with 7μF & 3μF So
$V_{3.9\mu F} = 20$ Volt
$q_{3.9\mu F} = CV = 3.9\mu \times 20 = 78\ \mu C$
Net charge flow from E, 78μC + 42μC = 120μC

$$V_{12\mu F} = \frac{Q}{C} = \frac{120}{12} = 10\ \text{Volt}$$

KVL,
+E – 10 – 20 = 0
E = 30 Volt
Final Answer: (1)

21) **Explaining:** Find the value of equivalent resistance
Concept: Series and parallel combination of resistance
Solution:

R_{ab} (Between a and b) : The upper series paths reduce to $\frac{23}{2}\Omega$, which is in a parallel with the 6Ω resistor :

$$R_{ab} = \frac{6 \times \frac{23}{2}}{6 + \frac{23}{2}} = \frac{138}{35}\Omega$$

R_{db} (Between d and b): This forms a **balanced Wheatstone bridge** $\left(\frac{2}{3} = \frac{6}{9}\right)$, so the central 5Ω resistor is removed. The parallel branches (8Ω and 12Ω) give:

$$R_{db} = \frac{8 \times 12}{8 + 12} = \frac{24}{5}\ \Omega$$

R_{ac} (Between a and c): Three paths ($5\Omega, 5\Omega,$, and 15Ω) are in parallel:

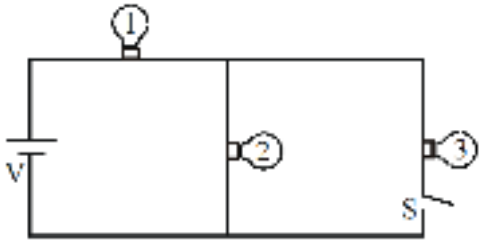
$$\frac{1}{R_{ac}} = \frac{1}{5} + \frac{1}{5} + \frac{1}{15} \Rightarrow R_{ac} = \frac{15}{7}\ \Omega$$

R_{dc} (Between d and c): The network reduces to a 3Ω resistor in parallel with a $\frac{23}{4}\ \Omega$ branch:

$$R_{dc} = \frac{3 \times \frac{23}{4}}{3 + \frac{23}{4}} = \frac{69}{35}\ \Omega$$

Final Answer: (1)

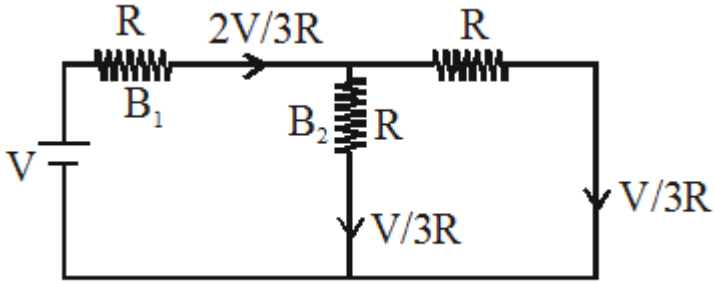
22)



Initially $P_1 = P_2$ & $P_3 = 0$

$P_{system} = \frac{V^2}{R_{eq}} = \frac{V^2}{2R}$

When switch closed,



$R_{eq} = \frac{R}{2} + R = \frac{3R}{2}$

Since, $R_{eq} \downarrow$ overall $I\uparrow$, $P_1\uparrow$ but $P_2\downarrow$ & $P_3\uparrow$.

Thus,

B_1 decreases, B_2 increases & B_T decreases

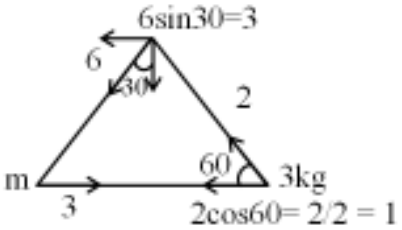
23) **Explain Question:**

We have to calculate value of m

Concept:

Motion of centre of mass

Solution:



$v_{cm}x = \frac{m \times 3 - 3(1) - 1 \times 3}{4 + m} = 0$

3m = 6

m = 2

Final Answer: Option (2).

24)

Explain question:
We have to calculate potential energy of rod.

Concept:
Centre of mass of continuous system.

Solution:



$dU = dmgx$

$dU = \lambda dxgx$

$$\int_0^U dU = \int_{x=0}^{x=1} (2x+2)xg\,dx$$

$$\int_0^U dU = g\int_{x=0}^{x=1} (2x^2+2x)\,dx$$

$$\int_0^U dU = g\left[\frac{2x^3}{3}+x^2\right]_0^1$$

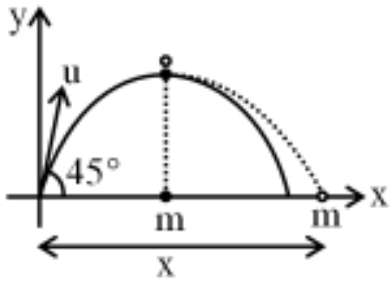
$$\int_0^U dU = g\left[\left\{\frac{2(1)^3}{3}+(1)^2\right\}-\{0+0\}\right]$$

$$\int_0^U dU = g\left[\frac{2}{3}+1\right]$$

$$U = \frac{5}{3}g$$

Final answer: Option (1)

25) **Explain Question :** A bomb explode at highest point into two equal parts first part comes at rest than range of second part is.
Concept : Conservation of linear momentum.
Solution :



$$R = \frac{u^2\sin(2\theta)}{g} = 40m$$

The COM will hit the ground at distance R.

$$dx_{com} = \frac{m_1dx_1 + m_2dx_2}{m_1+m_2}$$

$$R = \frac{m\left(\frac{R}{2}\right) + mx}{2m}$$

$$40 = \frac{(40/2) + x}{2}$$

$80 = 20 + x \Rightarrow x = 60$ m

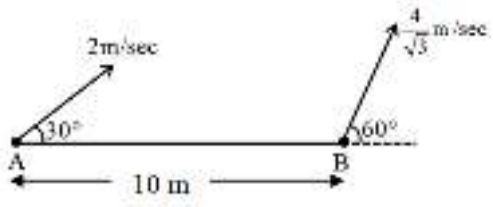
Final answer : Option (2)

26)

Question Explanation : Two particles A and B start from ground points 10 m apart. Their initial speeds and launch angles are given. Find the instantaneous angular speed of B about A at launch.

Concept : This question is based on relative angular speed

Solution :



$$\omega_{rel} = \frac{\text{relative velocity perpendicular to line joining}}{\text{separation between them}}$$

$$\omega_{rel} = \frac{\frac{4}{\sqrt{3}}\sin 60^o - 2\sin 30^o}{10} = \frac{\frac{4}{\sqrt{3}}.\frac{\sqrt{3}}{2} - 2.\frac{1}{2}}{10}$$

$$\omega_{rel} = \frac{2-1}{10} = 0.1 \text{ rad/sec}$$

Final answer : Option (1)

27)

Explain Question : How much the maximum safe speed increases if road radius becomes four times but banking angle remains same

Concept : Banking of road
Formula : $v = \sqrt{Rg\tan\theta}$

Solution :

$$v \propto \sqrt{R}$$

$$\frac{v_2}{v_1} = \sqrt{\frac{4R}{R}} = 2$$

Percentage increase

$$= \frac{v_2 - v_1}{v_1} \times 100 = \left(\frac{v_2}{v_1} - 1\right) \times 100 = (2-1) \times 100 = 100\%$$

Final answer : Option (4)

28) **Explaining :** To find the correct graph for the situation.
Concept : This question is based on concept of potential energy.
Solution : $F_x = -\frac{dU}{dx}$

$U = \frac{1}{2} kx^2 \Rightarrow F = -kx$

$\frac{1}{2}kx_{max}^2 = U_{max} \Rightarrow \frac{1}{2}(kx_{max})\, x_{max} = U_{max}$

$\Rightarrow \frac{1}{2}(F_{max})(10) = 15$

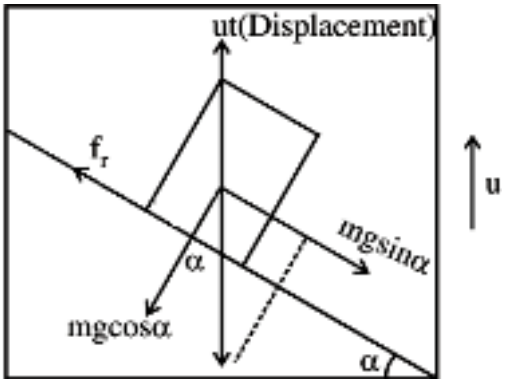
$\Rightarrow F_{max} = 3N$

Final Answer : (1)

29) **Question Explanation:** Evaluate statements about zero work done and work done by friction in a system.

Concept: Work Done by Frictional Force.

Solution:
Assertion: No work is done by a force on an object when the point of application of the force does not move. (True, S = 0).
Reason: Work done by friction on a block not sliding in an elevator moving at constant velocity is $W = mgut\sin^2\alpha$. is also true becouse,



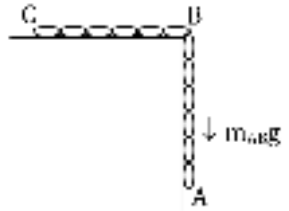
As block is not sliding
$f_r = mg\sin\alpha$
Work done by friction on the block w.r.t ground
$W = f_r ut\cos(90° - \alpha)$
$W = (mg\sin\alpha)(ut)\sin\alpha$
$= mgut\sin^2\alpha$

Final Answer : Option (2)

30)

Explain Question : A machine of 75% efficiency uses 12 J energy to lift a 1 kg mass. Find the velocity of the mass when it falls through the same height.
Concept : This question is based on efficiency, work - energy theorem, and law of conservation of energy.
Solution :
Given : n = 75% = $\frac{75}{100}$, E = 12J, m = 1kg
Useful work (gain in potential energy) :
mgh = n × E = $\frac{75}{100} \times 12 = 9J$
When the mass falls through the same height, by law of conservation of energy :
$\frac{1}{2}mv^2 = mgh = \frac{75}{100} \times 12$
$\Rightarrow v = \sqrt{\frac{2 \times 75 \times 12}{100 \times m}}$
Substitute m = 1 kg.
$v = \sqrt{\frac{3}{4} \times 24} = \sqrt{18} m/s$
Final answer : 2

31) **Explaining :** To find work done to pull the hanging part of chain.
Concept : This question is based on concept of work done.
Solution :



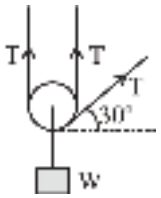
$m_{AB} = \frac{2}{5}m$
$AB = \frac{2\ell}{5}$
$W = m_{AB}g \times \frac{AB}{2} = \frac{2mgl}{25}$
Final Answer : (1)

32) **Explaining :** To find the condition for equilibrium of the body.
Concept : This question is based on concept of equilibrium of body.
Solution : $\vec{F}_1 + \vec{F}_2 = 0$
$(1-t)\hat{i} + 2(t^2-1)\hat{j} = \vec{0}$
⇒ t = 1s
Final Answer : (2)

33) **Explaining :** To find the impulse when string just becomes tight.
Concept : This question is based on concept of impulse momentum theorem.
Solution : I = ΔP = m(v–u) = $2\left(\sqrt{2gh} - 0\right)$

$= 2\sqrt{2 \times 10 \times 5}$
= 20 N–S
Final Answer : (3)

34) **Explaining :** To calculate the value of weight w.
Concept : This question is based on concept of equilibrium.
Solution :



T = 60
W = T + T + T sin 30º

$= 60 + 60 + 60 \times \frac{1}{2} = 150$ N
Final Answer : (3)

35) **Explaining :** To find the friction force vector.
Concept : This question is based on concept of friction force in a plane.
Solution :
$\vec{f}_r$ is antiparallel to $\hat{u}$
$\vec{f}_r = \mu_k N = 0.5 \times 5 \times 10 = 25N$

$$\vec{f}_r = -25 \times \frac{(\hat{i}+\hat{j})}{\sqrt{2}} = \frac{-25}{\sqrt{2}}(\hat{i}+\hat{j})$$

Final Answer : (3)

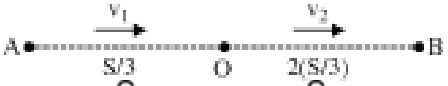
36) **Explaining :** To find the pseudo force on person when viewed from different frame of refrence.
Concept : This question is based on concept of frame of refrence.
Solution :
Pseudo force from ground = Zero (inertial frame)
Pseudo force from platform = $-m_{body} \cdot \vec{a}_{frame}$

$= -50(-5\hat{i}) = 250\hat{i}$
Final Answer : (2)

37) **Explaining :** To find a-t graph from given v-t graph.
Concept : This question is based on concept of conversion of graph.
Solution :
 v = $10\, e^{-t/2}$
 a = $\frac{dv}{dt} = 10e^{-t/2} \times \left(\frac{-1}{2}\right) = -5e^{-t/2}$
at t = 0, a = – 5 m/s^2
and as t increases, a decreases exponentially in negative.
Final Answer : (1)

38) **Explaining :** To find the average acceleration for the given period of motion.
Concept : This question is based on concept of average acceleration.
Solution : $\vec{a}_{avg} = \frac{\vec{v}_f - \vec{v}_i}{t} = \frac{2\hat{i} + 2\sqrt{3}\hat{j} - 2\hat{i}}{2} = \sqrt{3}\,\hat{j}$
$\sqrt{3}\,m/s^2,\ \text{along } y-\text{axis}$
Final Answer : (2)

39) **Explain the Question :** Asking about the average speed for given journey.
Concept : This question is based on average speed.
Solution :



$v_{av} = \frac{S}{t_1 + t_2} = \frac{S}{\left(\frac{S/3}{v_1}\right) + \frac{2S}{3}\Big/ v_2}\ v_{av} = \frac{3v_1v_2}{2v_1 + v_2}$
Final Answer : 2

40)

Question Explanation:
The question asks to final the horizontal velocity of the ball so that ball can reach from one building to another.
Concept:
Horizontal projection projectile motion from a height
Solution:
In verticle direction.
$h = 150 - 27.5 = 122.5m$
$t = \sqrt{\frac{2h}{g}}$

$t = \sqrt{\frac{2 \times 122.5}{9.8}} = 5$ sec
Hence, $u = \frac{R}{t}$ (in horizontal direction)
$u = \frac{30}{5} = 6m/s$
Final answer: Option (2)

41) **Explaining :** To find the maximum possible error in value of g.
Concept : This question is based on concept of error.
Solution : $T = 2\pi\sqrt{\frac{\ell}{g}} \Rightarrow g = 4\pi^2\frac{\ell}{T^2}$
$\frac{\Delta g}{g} \times 100 = \frac{\Delta \ell}{\ell} \times 100 + \frac{2\Delta T}{T} \times 100$
$= \frac{0.1}{100} \times 100 + 2 \times \frac{0.1}{50} \times 100$
= (0.1 + 0.4) %
= 0.5 %

Final Answer : (1)

42) **Explaining :** To find the correct order of percentage error.
Concept : This question is based on concept of percentage error in R, S, P and D.
Solution : $\frac{\Delta P}{P} = \pm\left(\frac{\Delta x}{x} + \frac{\Delta y}{y}\right)\frac{\Delta P}{P} \times 100 = \pm\left(\frac{0.6}{12.0} + \frac{0.2}{5.0}\right) \times 100$
$= \pm (5 + 4) = \pm 9\%$
$\frac{\Delta D}{D} = \pm\left(\frac{\Delta x}{x} + \frac{\Delta y}{y}\right) = \pm 9\%$
$R = x + y$, $R \pm \Delta R = 17.0 \pm 0.8$
$\frac{\Delta R}{R} \times 100 = \pm\frac{0.8}{17.0} \times 100 = \pm 4.7\%$
$S = x - y$, $S \pm \Delta S = 7.0 \pm 0.8$
$\frac{\Delta S}{S} \times 100 = \pm\frac{0.8}{7.0} \times 100 = \pm 11.4\%$
correct order of % error is
$R < P = D < S$
Final Answer : (3)

43) **Explain Question:** Determine the dimensional formula of mass using energy (E), velocity (V), and force (F) as fundamental units.

Concept: Dimensional Analysis.

Solution:
Energy $E = \frac{1}{2}mv^2$
In terms of dimensions: $[E] = [M][v]^2$
$[M] = \frac{[E]}{[v]^2}$
$[M] = [Ev^{-2}]$

Final Answer: Option (3)

44) **Explaining :** To find the diameter of the spherical object.
Concept : This question is based on concept of vernier calliper.
Solution : 1 M.S.D. = 0.1 cm (Given)
10 V.S.D. = 9 M.S.D. ⇒ 1 V.S.D. = 0.9 M.S.D.
1 V.S.D. = 0.9 × 0.1 cm = 0.09 cm
L.C. = 1 M.S.D. – 1 V.S.D.
= (0.1 – 0.09) cm = 0.01 cm
Reading = 5.08 cm
Positive zero error = 0.1 cm (Given)
 Final Corrected reading = (5.08 – 0.1) cm = 4.98 cm
Final Answer : (3)

45)

Explain the Question : A screw gauge with 50 circular divisions and a pitch of 0.5 mm is used to measure a wire. The main scale reading (MSR) is 3.5 mm, the circular scale reading (CSR) is 32, and there is a positive zero error of 0.06 mm. calculate the correct diameter of the wire.
Concept : This question is based on measurement using a micrometer screw gauge.
Formula :

$$\text{Least count(LC)} = \frac{\text{Pitch}}{\text{No. of circular divisions}}$$

Measured reading = Main scale reading + (Circular scale reading × LC) – Zero error.
Solution : Pitch = 0.5 mm
Circular divisions = 50

$$(LC) = \frac{0.5}{50} = 0.01\text{mm}$$

Measured reading = 3.5 + (32 × 0.01) – 0.06 = 3.5 + 0.32 – 0.06 = 3.76 mm
Final answer : 2

CHEMISTRY

46) **Explaining :** The number of atoms of Cr and O are 4.8×10^{10} and 9.6×10^{10} respectively. Its empirical formula is –
Concept : This question is based on empirical formula.
Solution :
E. f. ⇒ obtain from simple ratio of atom
Gives Cr = 4.8×10^{10} , O = 9.6×10^{10}
Simplest ratio 1 : 2
Empirical formula = CrO_2
Final Answer : (2)

47) **Explaining :** If 0.5 mole of $BaCl_2$ is mixed with 0.2 mole of Na_3PO_4 the maximum number of moles of $Ba_3(PO_4)_2$ that can be formed is -
$3BaCl_2 + 2Na_3PO_4 \rightarrow Ba_3(PO_4)_2 + 6NaCl$
Concept : This question is based on L.R.
Solution :

$$3\,BaCl_2 + 2Na_3PO_4 \rightarrow Ba_3(PO_4)_2 + 6NaCl$$

$$\downarrow \qquad \downarrow \qquad \downarrow$$

3 moles 2 moles 1 mol
Given 0.5 moles 0.2 moles
so 0.3 moles of $BaCl_2$ will react with
0.2 moles of Na_3PO_4 and gives 0.1 mol of $Ba_3(PO_4)_2$
Final Answer : (4)

48) **Explaining :** The quantum number of 20th electron of Fe(Z = 26) would be :–
Concept : This question is based on quantum number.
Solution :
$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^6$
n = 4, □ = 0, m = 0
s = $+\frac{1}{2}$, $-\frac{1}{2}$

Final Answer : (3)

49) **Explaining :** Wavelength of first line of Lyman series in terms of R of hydrogen atom :-
Concept : This question is based on Rydberg formula.
Solution :
First line of Lyman 2 → 1

$$\frac{1}{\lambda} = R(1)^2\left(\frac{1}{(1)^2} - \frac{1}{(2)^2}\right)$$

$$= R\left(\frac{1}{1} - \frac{1}{4}\right)$$

$$\frac{1}{\lambda} = \frac{3R}{4}$$

$$\Rightarrow \lambda = \frac{4}{3R}$$

Final Answer : (2)

50)

Explaining : Equilibrium constant (K) of the reaction.
Concept : This question is based on mode of representation of reaction
Solution :
Reverse (I)
ADD in (II)
Multiply (III) × 3

$$\frac{1}{K_1} \times K_2 \times (K_3)^3$$

Final Answer : (4)

51)

Explaining : D.O.D.
Concept : This question is based on vapour density and D.O.D.
Solution :

$$\alpha = \frac{D-d}{(n-1)d}$$

$$n = \frac{5}{2} \qquad D = \frac{M.W_A}{2} = \frac{48}{2} = 24$$

$$\alpha = \frac{24-16}{\left(\frac{5}{2}-1\right)\times 16} = \frac{8}{\frac{3}{2}\times 16} = \frac{1}{3}$$

Final Answer : (1)

52)

Explaining : degree of hydrolysis
Concept : This question is based on hydrolysis of W.A + WB
Solution :
WAWB salt

h → independent of C
Final Answer : (1)

53) **Explaining :** Buffer solution
Concept : This question is based on buffer solution
Solution : $(\text{gm eq.})_{WA} > (\text{gm eq.})_{SB}$
Final Answer : (2)

54)

Explaining : pH of the solution
Concept : This question is based on Buffer Solution
Solution :
[Salt] = 0.1 M, [Acid] = 0.1 M

$K_a = 1.8 \times 10^{-5}$; $pH = -\log K_a + \log \frac{[Salt]}{[Acid]}$

$= -\log 1.8 \times 10^{-5} + \log \frac{0.1}{0.1} = -\log 1.8 \times 10^{-5}$

pH = 4.75
Final Answer : (3)

55)

Explaining : pH
Concept : This question is based on solubility
Solution :
$Cu(OH)_2 \rightarrow AB_2$ type salt

$K_{sp} = 4s^3 = 3.2 \times 10^{-17}$

$4s^3 = 32 \times 10^{-18}$

$s = 2 \times 10^{-6}$

$[OH^-] = 2s = 4 \times 10^{-6}$

pOH = 6 – log4

= 5.4

pH = 8.6
Final Answer : (4)

56)

Explaining : Standard heat of trimerisation
Concept : This question is based on heat of formation
Solution :

$\Delta H_{reaction} = \Delta H^o_{f(C_6H_6)} - 3\Delta H^o_{f(C_2H_2)}$

= 85 – 3(230)

= –605 kJ mol^{-1}
Final Answer : (1)

57)

Explaining : Enthalpy change in KJ
Concept : This question is based on enthalpy of reaction.
Solution :

$$3C(s)+\frac{3}{2}O_2(g)\longrightarrow 3CO(g);\qquad \Delta H=-330kJ$$

$$Cr_2O_3(s)\longrightarrow 2Cr(s)+\frac{3}{2}O_2(g);\Delta H=+1130kJ$$

On adding, $3C(s)+Cr_2O_3(s)\longrightarrow 2Cr(s)+3CO(g);\quad \Delta H=800KJ$

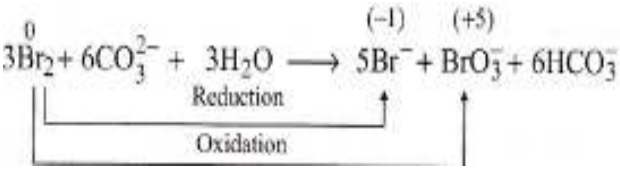
Final Answer : (2)

58)

Explaining : Disproportionation reaction
Concept : This question is based on disproportionation reaction.
Solution : $\overset{+4}{Mn}O_2 + 4H\overset{-1}{Cl} \rightarrow \overset{+2}{Mn}Cl_2 + \overset{0}{Cl_2} + 2H_2O$
Final Answer : (3)

59)

Explaining : Following reaction
Concept : This question is based on balancing of redox reaction
Solution :



It is disproportionation oxidation in which bromine (Br_2) is oxidised as well as reduced.
Final Answer : (3)

60) **Explaining :** Maximum voltage.
Concept : This question is based on E.M.F. of cell.
Solution :

$$Zn+2Ag^{+}\longrightarrow 2Ag+Zn^{2+};$$

$$E^0_{cell}=+0.76+0.80=+1.56V$$

Final Answer : (2)

61) **Explaining :** E^0_{cell}
Concept : This question is based on E^0_{cell} and $\Delta G°$
Solution : $\Delta G^\circ < 0$ for spontaneous $Rx^n; E^\circ_{cell} > 0; k > 1$
Final Answer : (1)

62) **Explaining :** Cell potential.

Concept : This question is based on Nernst equation.
Solution :

$$E = E^{\circ} - \frac{0.059}{4}\log\left\{\frac{\left[Fe^{2+}\right]^2}{\left[H^+\right]^4\left[P_{O_2}\right]}\right\} = 1.57\ V$$

Final Answer : (4)

63) **Explaining :** Vapour pressure of the two pure liquids x and y.
Concept : This question is based on Raoult's Law.
Solution : -
Molar ratio of 1 : 1 means $x_X = x_Y = 0.5$

$p_{total} = x_X p_X^o + x_Y p_Y^o$

$\Rightarrow 400 = 0.5p_X^o + 0.5p_Y^o$

or $p_X^o + p_Y^o = 800$ ....(i)

Molar ratio of 1 : 2 means, $x_X = \frac{1}{3},\ x_Y = \frac{2}{3}$

$\Rightarrow 350 = \frac{1}{3}p_X^o + \frac{2}{3}p_Y^o$

or $p_X^o + 2p_Y^o = 1050$ ...(ii)

Eqn. (ii) – Eqn. (i) gives $p_Y^o = 250\ mm$

Putting in (i), we get $p_X^o = 550\ mm$

Final Answer : (4)

64)

Explaining : Rate Law
Concept : This question is based on rate law
Solution :
Keeping [A] constant when [B] is halved, rate becomes half.
= Order w.r.t B = 1 when [A] and [B] both are doubled, rate becomes 4 times
= Order w.r.t. A = 1
Final Answer : (2)

65)

Explaining : rate constant
Concept : This question is based on first order reaction
Solution : $k = \frac{1}{5}\ln\left(\frac{100}{30}\right) = 0.24\ min^{-1}$
Final Answer : (2)

66)

Explaining : Which is correct
Concept : This question is based on arrhenius equation
Solution : Rate constant increases exponentially with decreasing activation energy and increasing temperature.

Final Answer : (4)

67)

Explaining : Half-life for this reaction
Concept : This question is based on first order reaction
Solution : $t_{1/2} = \frac{\ln 2}{0.0541} = 12.8\,\text{sec}$
Final Answer : (2)

68)

Explain Question: Match the relationship Van't Hoff factor (i) and the nature of the solute.
Concept: This question is based on Van't Hoff factor.
Solution:
Dissociation $\Rightarrow i > 1$; (I – a)
Association $\Rightarrow i < 1$; (II – b)
Non electrolyte $\Rightarrow i = 1$; (III – d)
For electrolyte $\Rightarrow i \neq 1$, $i < 1$ or $i > 1$; (IV – c)
Final Answer: (4)

69) **Question Explanation:** We need to check the correctness of two statements related to the equilibrium constant.
Concept: This question is based on Factors affecting the Equilibrium constant (K).
Solution

A. Addition of inert gas does not change equilibrium constant.

B. Equilibrium constant (K) depends only on temperature, not on pressure, volume, or concentration.

Reasoning

A. Statement I: Correct (no effect of inert gas on k).

B. Statement II: Correct (K changes only with temperature).

Final Answer: (4)

70)

Question Explanation: Assertion says $\Delta H_f°$ for $H_2(g)$ and $Cl_2(g)$ is zero; Reason says $\Delta H_f°$ for every element in its stable standard state is zero.
Concept: This question is based on standard enthalpy of formation.
Solution:
By definition, standard enthalpy of formation of an element in its most stable form (standard state) at 1 atm and 298 K is taken as zero.
$H_2(g)$ and $Cl_2(g)$ are elements in their stable standard states, so their $\Delta H_f° = 0$. Hence both statements are true and (R) correctly explains (A).

Final Answer: (4)

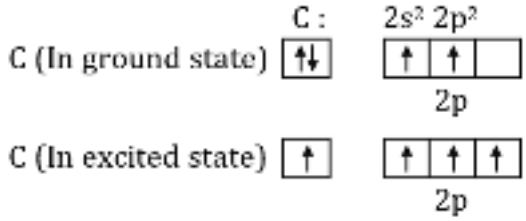
71) **Explaining :** When the substances SiC, KCl, CH_3OH and C_2H_6 are arranged in order of increasing melting point, what is the correct order ?
Concept : This question is based on melting point of compounds
Solution :

C_2H_6	<	CH_3–OH	<	KCl	<	SiC
(Molecular lattice)		(Molecular lattice) (with H-bonding)		(Ionic lattice)		(Covalent lattice)

Final Answer : (4)

72) **Explaining :** In which of the following transformation, the bond order has increased and the magnetic behaviour has changed ?
Concept : This question is based on Bond order
Solution :
During $C_2^+ \rightarrow C_2$ formation, e^- is added in BMOs so, bond order increase and as e^- is added in π_{2p} so, all electrons are paired, hence diamagnetic.
Final Answer : (1)

73) **Explaining :** Assuming pure 2s and 2p orbitals of carbon are used in forming CH_4 molecule, which of the following statement is false ?
Concept : This question is based on VBT
Solution :

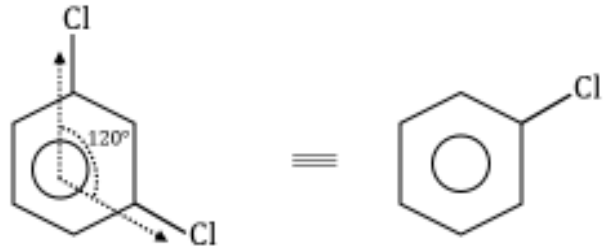


In the unhybridized state of carbon, 2p orbitals are 90° to one another and each one will overlap with 1s orbital of three hydrogen atoms, thus three C–H bonds are formed which are 90° to one another. For the fourth hydrogen atom, its 1s orbital may overlap with non-directional 2s orbital of the carbon and this σ-bond will be weaker than σC–H bonds formed by 2p-1s overlap. In such situation CH_4 molecule can never has tetrahedral geometry.
Final Answer : (3)

74) **Explaining :** Match the Following -
Concept : This question is based on bond characteristics
Solution : A → 1, B → 2, C → 3, D → 4
Final Answer : (1)

75) **Explaining :** Which of the following orbital combination can not form π-bond ?
Concept : This question is based on overlapping
Solution : $d_{x^2-y^2} + p_y$ (side wise overlap) → Non-bonding
Final Answer : (2)

76) **Explaining :** Which of the following compounds has dipole moment approximately equal to that of chlorobenzene ?
Concept : This question is based on dipole moment
Solution :

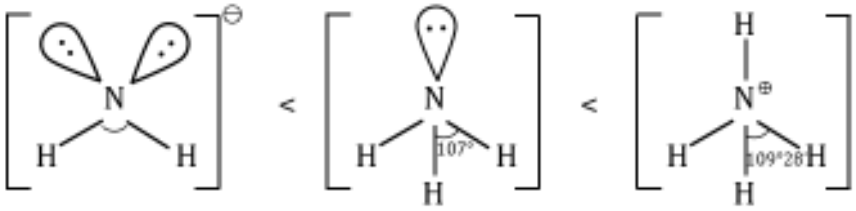


For Cl (–I > + M)

$$\mu_R = \sqrt{\mu^2 + \mu^2 + 2\mu^2 \cos\theta} = \sqrt{2\mu^2 + 2\mu^2 \cos 120^\circ}$$
$$= \sqrt{2\mu^2 - \mu^2} \Rightarrow \sqrt{\mu^2} = \mu$$

Final Answer : (2)

77) **Explaining :** The bond angle of NH_3, NH_4^+ and NH_2^- are in the order :
Concept : This question is based on bond angle
Solution :
When hybridization of central atom and surrounding atom are same but number of lone pairs are different, bond angle decreases with increase in number of lone pair.



Final Answer : (2)

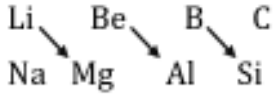
78) **Explaining :** The boiling points at atmospheric pressure of HF, H_2S, NH_3 can be arranged in the following order.
Concept : This question is based on boiling point order.
Solution :
Strength of H-bond per
Number of H-bond per HF → 2
Number of H-bond per NH_3 → 2
H_2S does not show hydrogen bonding.
Order of boiling point –– HF > NH_3 > H_2S.
Final Answer : (1)

79) **Explaining :** In terms of polar character, which of the following order is correct ?
Concept : This question is based on polarity
Solution : As electronegativity difference of bond increases, polar character also increases.
Final Answer : (2)

80) **Explaining :** Analyse both the statements
Concept : This question is based on MOT and its application.
Solution :
N_2 and NO^+ both are diamagnetic substances as both have zero unpaired electron. NO^+ is isoelectronic with N_2.
Final Answer : (1)

81) **Explaining :** Which of the following has maximum solubility in the below mentioned alkaline earth metal sulphates ?
Concept : This question is based on solubility of alkaline earth metal sulphates.
Solution :
Solubility order : $BeSO_4 > MgSO_4 > CaSO_4 > SrSO_4 > BaSO_4$
Solubility ∝ Hydration energy
Hydration energy decreases down the group (II-A)
Final Answer : (4)

82) **Explaining :** The diagonal partner of element B is :
Concept : This question is based on diagonal relationship
Solution :
Element of second period show similarity in properties with the elements of third period that are placed diagonally opposite. This type of similarity is called diagonal relationship, e.g.,



Final Answer : (3)

83) **Explaining :**
The element Z = 114 has been discovered recently. It will belong to which of the following family/group and electronic configuration?
Concept : This question is based on finding group, period and block of an element.
Solution :
The electronic configuration of the element with Z = 114 is [Rn] $5f^{14}$ $6d^{10}$ $7s^2$ $7p^2$. Hence it belongs to the carbon family which has the same outer electronic configuration.
Final Answer : (1)

84) **Explaining :** The first ionisation enthalpies of Na, Mg, Al and Si are in the order:
Concept : This question is based on ionization enthalpy
Solution :
Na < Mg : Moving from Sodium ($1s^2 2s^2 2p^6 3s^1$) to Magnesium ($1s^2 2s^2 2p^6 3s^2$), the ionization energy increases due to an increase in nuclear charge and the removal of an electron from a stable, fully-filled 3s orbital.
Mg > Al : Although Aluminum has a higher nuclear charge than Magnesium, its first ionization enthalpy is lower (577 kJ/mol vs 737 kJ/mol). This is because Aluminum ([Ne]$3s^2 3p^1$) has its outermost electron in a 3p orbital, which is further from the nucleus and more shielded than the 3s orbital of Magnesium. Additionally, Magnesium has a stable, fully-filled subshell.
Al < Si : Moving from Aluminum to Silicon ([Ne]$3s^2 3p^2$), the ionization energy increases again due to the significant increase in effective nuclear charge and decreasing atomic size

Final Answer : (1)

85) **Explaining :** Which of the following groups of elements is assigned zero electronegativity :
Concept : This question is based on electronegativity.
Solution : Noble gases have fully filled electronic configuration thus electronegativity is assigned to be zero.
Final Answer : (2)

86) **Explaining :** Which of the following can readily react with NaOH solution :
Concept : This question is based on nature of oxide
Solution : Acidic oxide will react with NaOH readily.
Final Answer : (4)

87) **Explaining :** If the ionic radius of K^+ and F^- are nearly same 1.34 Å, then the atomic radii of K and F respectively will be :
Concept : This question is based on atomic radius
Solution :
$r_+ < r_{\text{parent atom}}$
$r_- > r_{\text{parent atom}}$
Final Answer : (3)

88) **Explaining :** Which of the following can't exist :
Concept : This question is based on existence and non-existence
Solution : Highest covalency of B is 4 and Al is 6.
Final Answer : (3)

89) **Explaining :** The ___ of any given element is not constant, it varies depending on the element to which it is bound :
Concept : This question is based on electronegativity
Solution : The electronegativity of any given element is not constant, it varies depending on the element to which it is bound.
Final Answer : (3)

90) **Explaining :** IUPAC name of the element with atomic number as 117 is :
Concept : This question is based on IUPAC naming
Solution : 117 - Ununseptium
Final Answer : (4)

BIOLOGY

91) **Solution :** A–1–i, B–2–ii, C–3–iii

Column I (Phylum)	Column II (Special Characters)	Column III (Example)
A. Porifera	1. Canal system	i. Sycon
B. Cnidaria	2. Cnidoblasts	ii. Physalia
C. Annelida	3. Metamerism	iii. Nereis

Final Answer : (1)

92) **Solution :** 1,2,3, and 5 are correct.
4 incorrect arthropods are triploblastic and coelomate.
Final Answer : (3)

93) **Solution :** A–2, B–3, C–1

Class	Character
A. Insecta	2. Three pairs of legs
B. Arachnida	3. No antennae
C. Crustacea	1. Two pairs of antennae

Final Answer : (1)

94) **Solution :** Pleurobrachia is an example of Ctenophora.
Final Answer : (2)

95) **Solution :** Cutaneous respiration enables amphibians to live in water also
Final Answer : (1)

96) **Solution :** A–3, B–2, C–1, D–4

Column I	Column II
A. Ampulla	3. Site of fertilisation
B. Isthmus	2. Narrow part of oviduct
C. Endometrium	1. Implantation
D. Cervix	4. Lower part of uterus

Final Answer : (3)

97) **Solution :** 1,2,3,5 correct, ovulation occur on the 14th day, not in the luteal phase
Final Answer : (3)

98) **Solution :** Fertilisation → Cleavage→ Implantation→ Gastrulation
Final Answer : (3)

99) **Solution :** Human placenta is a temporary endocrine tissue and secretes many hormones.
Final Answer : (1)

100) **Solution :** Zona pellucida develops a fertilisation membrane after entry of first sperm which prevents polyspermy.
Final Answer : (1)

101) **Solution :** 1, 2 and 3 are correct, Progesterone level is low during follicular phase. **Final Answer : (3)**

102) **Solution :** 1, 2, 3 and 4 are correct, In vasectomy vas deferens is cut and tied.
Final Answer : (3)

103) **Solution :** Progesterone + estrogen pills inhibit ovulation.
Final Answer : (1)

104) **Solution :** Sex determination is misused for female foeticide
Final Answer : (2)

105) **Solution :** A–2, B–1, C–3, D–4

Column I (Tissue)	Column II (Function)
A. Squamous epithelium	2. Diffusion
B. Cuboidal epithelium	1. Secretion
C. Columnar epithelium	3. Absorption
D. Cardiac muscle	4. Involuntary contraction

Final Answer : (1)

106) **Solution :** 1,2,4,5 correct, Frogs can respire through lungs, skin and buccopharyngeal cavity.
Final Answer : (3)

107) **Solution :** Mouth → Crop→ Gizzard→ Midgut
Final Answer : (3)

108) **Solution :** Cardiac muscles perform rhythmic contraction because they
are involuntary in function rhythmic contraction is due to intrinsic ability of pace maker cells.
Final Answer : (2)

109) **Solution :** Three chambered heart leads to mixing of blood due to single ventricle.
Final Answer : (1)

110) **Solution :** 1, 2 & 4 are correct, Striated muscles are voluntary.
Final Answer : (3)

111) **Solution :** A–1, B–2, C–3, D–4

Column I	Column II
A. Alveoli	1. Exchange of gases
B. Diaphragm	2. Increases thoracic volume
C. Haemoglobin	3. Transport of oxygen

D. Pleura	4. Double layered membrane of lungs

Final Answer : (2)

112) **Solution :** 1, 3, 4 and 5 are correct, Expiration is a passive process.
Final Answer : (3)

113) **Solution :** The site of diffusion is between alveolar epithelium and capillary endothelium.
Final Answer : (3)

114) **Solution :** Trachea → Bronchi → Bronchioles → Alveoli
Final Answer : (1)

115) **Solution :** The rate of diffusion of a gas is directly proportional to its solubility in the medium. CO_2 is 25 times soluble than oxygen.
Final Answer : (1)

116) **Solution :** Dependence on partial pressure is demonstrated by the fact that high pO_2 promotes binding of oxygen with haemoglobin.
Final Answer : (1)

117) **Solution :** Statement 1, 2 and 4 are correct.
Residual volume cannot be measured with a spirometer.
Final Answer : (3)

118) **Solution :** A–4, B–2, C–3, D–1

Column I	Column II
A. Plasma	4. Fluid matrix
B. RBC	2. Transport of gases
C. WBC	3. Defence
D. Platelets	1. Clotting

Final Answer : (4)

119) **Solution :** Statement 1, 3 and 4 are correct. Statement 2 and 5 are incorrect.
Haemoglobin is present in RBCs, Lymph do not contain RBCs.
Final Answer : (2)

120) **Solution :** Atrial systole → Ventricular systole → Joint diastole
Final Answer : (2)

121) **Solution :** SA node spontaneously generates electrical impulses that initiates the heartbeat so it is called pacemaker of heart.
Final Answer : (1)

122) **Solution :** In double circulation blood passes through heart twice in one complete cycle

Final Answer : (1)

123) **Solution :** Statement 1, 3 and 4 are correct.
Veins have thin walls than arteries.
Final Answer : (3)

124) **Solution :** A–Q, B–P, C–R, D–S
A. Grasshopper → Q. XX–XO TYPE
B. Birds → P. Female heterogamety
C. Honey bee → R. Haplodiploidy
D. Human beings → S. XX–XY
Final Answer : (1)

125) **Solution :** For **Aa × Aa**, probability of **Aa = 1/2**
For **Bb × Bb**, probability of **bb = 1/4**
Combined probability = **1/2 × 1/4 = 1/8**
Final Answer : (3)

126) **Solution :** A male with:
47 chromosomes
One extra X chromosome (XXY)
Gynaecomastia
is characteristic of **Klinefelter's syndrome**, Karyotype = **47, XXY**

Final Answer : (3)

127) **Solution :** A trait that:

A. appears only in **males**
B. is transmitted from **father to all sons**

is characteristic of **Y-linked inheritance**, because the Y chromosome passes directly from father to son.

Final Answer : (3)

128) **Solution :**
Assertion (A) is **true**: DNA double helix maintains a constant diameter because a purine always pairs with a pyrimidine.
Reason (R) is also **true**: A–T has 2 hydrogen bonds, while G–C has 3 hydrogen bonds.
However, the number of hydrogen bonds is **not the reason** for the constant diameter of DNA.
Final Answer : (2)

129) **Solution :** In a polynucleotide chain, nucleotides are joined together by **phosphodiester bonds**.

These bonds form between:

A. the phosphate group of one nucleotide
B. and the sugar of the next nucleotide.

Without phosphate groups, the chain cannot form
Final Answer : (3)

130) **Solution :** Semiconservative replication in **chromosomes** was experimentally proved by **Taylor** using autoradiography in *Vicia faba* root tip cells. (Meselson and Stahl proved it in DNA of bacteria.)
Final Answer : (2)

131) **Solution :**
A → Correct: VNTRs are a type of minisatellite DNA.
B → Incorrect: Even DNA from a single cell can be amplified using PCR for DNA fingerprinting.
C → Correct: DNA fingerprinting helps study population and genetic diversity.
D → Correct: Frederick Sanger developed methods for determining amino acid sequences in proteins.
Therefore, the correct statements are A, C and D.
Final Answer : (3)

132) **Solution :**
(1) Correct — mutations may cause loss or gain of gene function.
(2) Incorrect — large deletions/rearrangements are easy to comprehend.
(3) Correct — deletion of nucleotides can produce a **frameshift mutation** if the deletion is not in multiples of three.
(4) Correct — because the genetic code is comma-less, shifting the reading frame changes all downstream codons.
Final Answer : (2)

133) **Solution :** These anticodons pair with the mRNA codons:

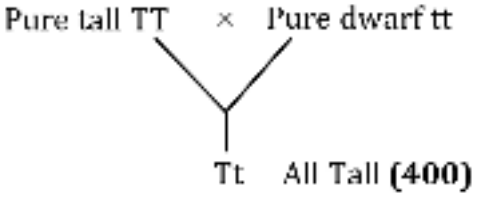
A. **AGU** pairs with **UCA**
B. **UAC** pairs with **AUG**

Now identify the amino acids coded by these codons:

A. **AGU → Serine (Ser)**
B. **UAC → Tyrosine (Tyr)**

Final Answer : (2)

134) **Solution :**



Final Answer : (3)

135) **Solution :** Tt × Tt cross results in 3 : 1 phenotypic ratio.
Final Answer : (3)

136) **Solution :** Whorl of carpel is gynoecium.
Final Answer : (1)

137) **Solution :** Both statements are independently true.
Final Answer : (2)

138) **Question Explanation :** The question asks to identify the incorrect statement regarding the dynamic process of pollen-pistil interaction.
Concept : Pollen-Pistil Interaction.
Solution: Pollen-pistil interaction includes all events from pollen deposition on the stigma until the pollen tube enters the ovule (not double fertilization). Double fertilization is a subsequent post-pollination event.
Final Answer : (4)

139) **Solution :** Castor is dicot, yet it has endosperm.
Final Answer : (4)

140) **Question Explanation :** The question is asking to identify the correct statements regarding taxonomy. **Concept :** Based on taxonomy and hierarchical categories. **Solution : Statement (A) → Incorrect:** Family has fewer similarities than genus and species. Genus and species are more specific, so similarities are greater at those levels. **Statement (B) → Correct:** Each rank (species, genus, family, etc.) is a unit of classification. Thus, only statement (B) is correct. **Final Answer : (2)**

141) **Concept :** This question is based on Classification systems.
Solution : Changes occurred because the criteria (e.g., morphology, anatomy, reproduction, molecular data) for classification evolved with new knowledge and technology.
Final answer : (1)

142) **Solution :** "Biology is the science of life forms and living processes. The living world comprises an amazing diversity of living organisms." Evolution explains this diversity, so biology can also be viewed as the story of life and its evolution on Earth.
Final Answer : (4)

143) **Solution :**
Assertion: True — All organisms (past, present, and future) are connected through the continuity of life and evolution.
Reason: True — All living organisms share common genetic material to varying degrees.
Final Answer : (1)

144) **Solution :**

(a) True - In the **Two-Kingdom Classification**, bacteria, blue-green algae, fungi, mosses, ferns, gymnosperms, and angiosperms were all placed in **Kingdom Plantae**.
(b) False - In the **Three-Domain System** (Archaea, Bacteria, Eukarya), **all eukaryotes are placed in a single domain — Eukarya**, not two domains.
(c) False - In **Whittaker's Five-Kingdom System**, **Chlamydomonas** and **Chlorella** are placed in **Kingdom Protista**, not Plantae.
Final Answer : (4)

145) **Solution :**
Assertion (A): True - Members of **Phycomycetes** are saprophytic, decomposers, parasitic and found on aquatic habitat.
Reason (R): The mycelium of Phycomycetes is **aseptate and coenocytic**, not septate.
Final Answer : (3)

146) **Solution :**
Monera: Mycoplasma, Archaebacteria.
Protista: Protozoa, Dinoflagellates.
Plantae: Mango, Wheat.
Final Answer : (1)

147)

Explain Question : The question asks to correctly match the kingdoms listed in Column I with their characteristic cell wall composition (Column II) and body organization (Column III)

Concept : This question is based on R.H. Whittaker

Solution :
2 (Protista) - a (Present in some) - iii (Cellular) → Protista have cell walls in some members and are mostly unicellular.
1 (Monera) - d (Polysaccharide + Amino acid) - iv (Cellular) → Monera have peptidoglycan (polysaccharide + amino acid) cell wall and are unicellular.
3 (Fungi) - b (Present chitin) - ii (Loose tissue) → Fungi cell wall chitin, loose tissue.
4 (Plantae) - c (Present cellulose) - i (Tissue/organ) → Plantae cell wall cellulose, well-developed tissue/organs.

Final Answer : option (1)

148) **Solution :** Grow in soil, logs, tree stumps, and as parasites**.**
Branched, septate mycelium.
Asexual spores generally absent.
Vegetative reproduction by fragmentation.
Sex organs absent; plasmogamy by fusion of vegetative cells.
Final Answer : (3)

149) **Solution :** Both Assertion and Reason are true. Viruses are non-cellular because they lack cellular organization and, outside the host cell, they do not perform metabolic activities, growth, or reproduction. Hence, the Reason correctly explains the Assertion.

Final Answer : (1)

150) **Solution:** Statement **(c)** is correct because basidiospores are produced **exogenously** and ascospores **endogenously**.
Statement **(d)** is also correct as some Basidiomycetes are called **bracket fungi**.
Statements **(a)** and **(b)** are incorrect.
Final Answer : (3)

151) **Solution :** Both statements are correct. **Yeast** is a unicellular fungus, and **dinoflagellates** possess two flagella—one transverse and one longitudinal.
Final Answer : (4)

152) **Solution:** In Phaeophyceae, food is stored as **carbohydrates (laminarin and mannitol)**, not proteins, so **(A)** is incorrect. Zoospores have **two unequal laterally attached flagella**, not equal flagella, so **(C)** is also incorrect.
Final Answer : (4)

153) **Solution:** Assertion is true because Rhodophyceae occur from well-lit regions to deep ocean waters. Reason is false because the dominant red pigment is **r-phycoerythrin**, not **r-phycocyanin**.
Final Answer : (4)

154) **Solution:** Both Assertion and Reason are true. Bryophytes are called the **amphibians of the plant kingdom** because they live on land but require water for sexual reproduction. Their role in plant succession on bare rocks/soil is true, but it does not explain why they are called amphibians.
Final Answer : (2)

155) **Solution: (A)** and **(E)** are correct. Mosses and lichens are pioneer species on bare rocks, and in gymnosperms both male and female gametophytes develop within sporangia present on the sporophyte. **(B)** is false (*Selaginella* is heterosporous), **(C)** is false (Cycas coralloid roots contain cyanobacteria, not mycorhiza), and **(D)** is false (main plant body of pteridophytes is sporophytic, not saprophytic). **Final Answer : (3)**

156) **Solution: Apical meristem** and **intrafascicular cambium** are primary meristems because they originate from embryonic meristematic tissue and contribute to primary growth.
Final Answer : (1)

157) **Explain Question -** The question asks to identify the incorrect statement related to plant anatomy and meristematic tissues.
Concept - Anatomy of flowering plants -complete
Solution -
Statement 1 is correct: Axillary buds arise from the shoot apical meristem.
Statement 2 is correct: In sunflower (dicot) stem, sclerenchymatous pericycle occurs above phloem in semi-lunar patches.

Statement 3 is incorrect: In a conjoint vascular bundle, phloem is located on the outer side of xylem, not inner.
Statement 4 is correct: Phloem parenchyma is absent in monocot stems.
Final answer - (3)

158) **Explain Question :**The question asks to identify the labels (A), (B), (C), and (D) shown in the four given diagrams
Concept : Tissue System - Epidermal Tissue System To Types Of Vascular Bundles.
Solution :
(A) = Guard cells
(B) = Radial condition
(C) = Conjoint closed
(D) = Conjoint open
Final answer : (4)

159) **Solution:** A **drupe** (e.g., mango, coconut) develops from a **monocarpellary superior ovary** and is typically **one-seeded**. Hence, **A -** Monocarpellary and **B -** One seeded.
Final Answer : (4)

160) **Solution:** Statements **A, B, and D** are correct. Statement **C** is incorrect because **gram and pea are dicots (two cotyledons)**, whereas **wheat and maize are monocots (one cotyledon)**; the statement has them reversed.
Final Answer : (3)

161) **Solution:** Both statements are true. In **bean, gram, and pea**, the endosperm is consumed during seed development, making them **non-endospermous seeds**. In dicot seeds, the **cotyledons are fleshy and store reserve food material**.
Final Answer : (1)

162) **Solution:** In monocot seeds, the **plumule is enclosed by the coleoptile** and the **radicle by the coleorrhiza**. The **micropyle is present above the hilum**.
Final Answer : (1)

163) **Solution:** Statement I is correct because a fruit is a **mature ripened ovary formed after fertilization**. Statement II is false because the scar on the seed coat is the **hilum**, whereas the **micropyle** is a small pore present above the hilum.
Final Answer : (2)

164) **Solution:** In the Cruciferae (Brassicaceae) family, flowers are **hypogynous**, **tetramerous**, and have **tetradynamous stamens**. Statement **(D)** is incorrect because the flowers are **actinomorphic (radially symmetrical)**, not zygomorphic.
Final Answer : (2)

165) **Solution:** The **Golgi apparatus** is called the **"traffic police" of the cell** because it modifies, packages, sorts, and transports proteins and lipids to their destinations.

Final Answer : (4)

166) **Solution:** Statement I is correct because **ATP synthase synthesizes ATP**. Statement II is incorrect because **mitochondria**, not chloroplasts, are the main site of cellular respiration.
Final Answer : (1)

167) **Solution:** Both statements are correct. **Schleiden and Schwann** formulated the cell theory, and **Rudolf Virchow** later expanded it by stating that **all cells arise from pre-existing cells**, giving the theory its final shape.
Final Answer : (2)

168) **Solution:** Assertion is true because ***Mycoplasma*** is the smallest known cellular organism. Reason is also true as bacteria are generally **3–5 µm** in length, but this does not explain why *Mycoplasma* is the smallest cell.
Final Answer : (2)

169) **Solution:** Statement (4) is correct. In a centriole, the **hub** in the proximal region is connected to the peripheral microtubule triplets by **proteinaceous radial spokes**. The other statements are incorrect.
Final Answer : (4)

170) **Solution :** I-c, II-d, III-b, IV-a
Pigments → Carotenoids, Anthocyanins (I–c).
Alkaloids → Morphine, Codeine (II–d).
Toxins → Abrin, Ricin (III–b).
Drugs → Vinblastin, Curcumin (IV–a).
Final Answer : (3)

171) **Question Explanation:** This question asks you to correctly match different biomolecules with their respective categories/functions.
Concept: Biomolecules → Functions of GLUT-4, collagen, insulin, trypsin, RuBisCO.
Explanation :

A. (a) GLUT-4 → transporter protein (ii)
B. (b) Collagen → intercellular ground substance (iv)
C. (c) Insulin → hormone (v)
D. (d) Trypsin → enzyme (iii)
E. (e) RuBisCO → most abundant protein (i)

Thus, the correct matching is: a–ii, b–iv, c–v, d–iii, e–i
Final Answer: (2)

172) **Solution:** Statements **(i), (ii), (iii), and (iv)** are correct. **(v)** is incorrect because cellulose is a **homopolymer** of glucose, and **(vi)** is incorrect because **glycogen is the storage polysaccharide of animals, while plants store starch**.
Final Answer : (2)

173) **Solution:** Statements **(a), (b), and (c)** are correct.
Statement **(d)** is incorrect because the most abundant protein in the biosphere is **RuBisCO**, not collagen.
 Number of correct statements = **3.**
Collagen in most abundant in animal kingdom.
Final Answer : (1)

174) **Solution:** Statements **A, B, and C** are correct. The protein part of a holoenzyme is the **apoenzyme**, while the non-protein part is the **cofactor**; prosthetic groups are organic cofactors tightly bound to the apoenzyme. Statement **D** is incorrect because many enzymes consist of **more than one polypeptide chain**.
Final Answer : (4)

175) **Question Explanation :** The question asks you to identify the correct stage of meiosis and match it with its specific event that occurs during that stage.



Concept : Celle cycle and cell division : Meiosis
Solution : The correct answer is Option 2. Anaphase I - Homologous chromosomes separate.
Explanation : The given diagram represents Anaphase I : The homologous Chromosomes separate, while sister chromatids remain associated at their centromeres.
Final Answer : (2)

176) **Solution:** Statement **B** is incorrect because **centromere splitting occurs during Anaphase-II**, not Anaphase-I. Statements **A, C, and D** are correct features of meiosis I.
Final Answer : (4)

177) **Solution:** A **bivalent** consists of two homologous chromosomes, each having two chromatids. Therefore, a bivalent contains **4 chromatids and 2 centromeres**.
Final Answer : (3)

178) **Solution: Lecithin** is a phospholipid containing glycerol, fatty acids, phosphate, and choline, whereas **triglyceride** consists of glycerol attached to three fatty acids.
Final Answer : (2)

179) **Explain Question:** Question asks about the class of enzymes that catalyse hydrolysis of ester, ether, peptide, and glycosidic bonds.
Concept: Classification of enzyme
Solution: Since the reactions mentioned involve hydrolysis of various bonds, the enzyme class involved is hydrolases.

Final answer: (3)

180) **Solution: Glucose, Fructose, Galactose, Maltose, and Lactose** are reducing sugars.
Sucrose and **Trehalose** are non-reducing sugars.
 Number of reducing sugars = **5.**
Final Answer : (3)